FrameNet II: Extended Theory and Practice

FrameNet II：扩展理论与实践

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Chapter 1 Introduction to the Project

第1章 项目介绍

1.0.1 Frame Semantics: Basic Concepts

1.0.1 框架语义学：基础概念

The theory of Frame Semantics grew out of the work of the Prof. Charles J. Fillmore, of the Linguistics Department at the University of California at Berkeley and his colleagues. 框架语义学理论源自加州大学伯克利分校语言学系查尔斯·J·菲尔莫教授及其同事的研究。

The theory asserts that people understand the meaning of words largely by virtue of the frames which they evoke. 该理论主张：人理解词义，很大程度依靠词语所唤起的语义框架。

The frames represent story fragments, which serve to connect a group of words to a bundle of meanings; 框架代表故事片段，它将一组词汇与一整套语义关联起来；

for example the term avenger evokes the Revenge frame, which describes a complex series of events and a group of participants: 例如，avenger（复仇者）会唤起"复仇(Revenge)"框架，该框架描述一整套复杂事件以及一组参与者：

the study of Frame Semantics attempts to define the frames and the "participants and props" involved in each of them. 框架语义学研究的目标，就是定义各个框架，以及框架所包含的"参与者与道具"。

The Revenge frame, discussed further in Sec. 3.1, is particularly complex; many frames are much simpler, such as the Placing frame (See Sec. 3.2.2.3), involving only an agent who does the placing, a theme (typically an inanimate object) that is placed somewhere, and a goal, the place where the theme winds up. 3.1节会进一步讨论的复仇框架尤其复杂；很多框架要简单得多，例如放置(Placing)框架（参见3.2.2.3节），仅包含执行放置动作的施事、被放置的主题（通常是无生命物体），以及目标——即主题最终所处的位置。

Usually, we can find the words that represent the fillers of these roles nearby in the text, often in the same sentence. 通常，我们可以在文本就近位置找到充当这些角色的词语，往往就在同一个句子之内。

Of course, the process of understanding an English sentence depends not only on knowing the words and the frames they evoke, but also understanding the grammatical constructions that determine the syntactic hierarchy of the sentence, and, in turn, the ordering of the words. 当然，理解英文句子，不仅要掌握词汇及其唤起的框架，还需要理解决定句子句法层级、进而决定词语语序的语法构式。

In parallel with the the work on Frame Semantics, Fillmore and his colleagues were also developing the theory of Construction Grammar, now being further developed by many syntacticians around the world. 与框架语义学研究同步，菲尔莫及其同事同时发展出构式语法理论，如今世界各地众多句法学家仍在拓展这套理论。

The work on the FrameNet project has always presupposed the existence of Construction Grammar, in which constructions not only define the relations between the frame-evoking elements and the role-filling elements, but also, in many cases, directly carry meaning themselves, which can also be represented as evoking a frame. FrameNet项目研究始终以构式语法为前提。在构式语法中，构式不仅定义唤起框架的成分与角色填充成分之间的关系；很多情况下构式本身直接承载语义，也可以被表征为唤起某个框架。

Since this document is intended mainly as a practical introduction to the work of the FrameNet project at the International Computer Science Institute, we will not discuss the theory of Frame Semantics (or Construction Grammar) further. 本文档主要是国际计算机科学研究所FrameNet项目实践导论，因此不再深入讨论框架语义学与构式语法理论。

For further information, with many references to the literature on Frame Sesmantics, please see Fillmore and Baker (2010), Petruck and de Melo (2014), and the FrameNet website, http://framenet.icsi.berkeley.edu. 如需更多资料与框架语义学相关文献，可参考Fillmore & Baker(2010)、Petruck & de Melo(2014)，以及FrameNet官网：http://framenet.icsi.berkeley.edu。

1.0.2 Frame Semantics and FrameNet

1.0.2 框架语义学与FrameNet

The Berkeley FrameNet project is creating an on-line lexical resource for English, based on frame semantics and supported by corpus evidence. 伯克利FrameNet项目基于框架语义学，依托语料证据，构建面向英语的在线词汇资源。

The aim is to document the range of semantic and syntactic combinatory possibilities– valences–of each word in each of its senses, through computer-assisted annotation of example sentences and automatic tabulation and display of the annotation results. 项目目标：借助计算机对例句进行标注，自动汇总展示标注结果，记录每个词汇各个义项全部的语义-句法组合可能性，即配价。

The major product of this work, the FrameNet lexical database, currently contains more than 13,000 lexical units (defined below), about 7,000 of which are fully annotated, in more than 1,000 hierarchically-related semantic frames, exemplified in more than 200,000 annotated sentences.1 项目主要产出是FrameNet词汇数据库：目前包含13000余个词汇单元（定义见下文），其中约7000个完成完整标注；分布在1000多个具备层级关联的语义框架中，标注例句数量超过20万句[1]。

The database has gone through seven releases, and is now in use by hundreds of researchers, teachers, and students around the world. (See the FrameNet Users page on our web-site.) 数据库已经发布7个版本，全球数百位研究者、教师、学生正在使用（详见官网的FrameNet用户页面）。

Active research projects are producing comparable frame-semantic lexicons for other languages and to devise means of automatically labeling running text with semantic frame information. 已有多项研究项目，为其他语言构建同类框架语义词典，同时研究如何对连续文本自动标注语义框架信息。

[1]Daily status updates are available at https://framenet.icsi.berkeley.edu/fndrupal/current_status. [1]每日状态更新地址：https://framenet.icsi.berkeley.edu/fndrupal/current_status。

A lexical unit (LU) is a pairing of a word with a meaning. 词汇单元(LU)，是词语与其某一个义项的配对。

Typically, each sense of a polysemous word belongs to a different semantic frame, a script-like conceptual structure that describes a particular type of situation, object, or event along with its participants and props. 一般来说，多义词的每个义项分属不同语义框架；框架是脚本式概念结构，描述某一类场景、事物或者事件，以及对应的参与者、道具。

For example, the Apply heat frame describes a common situation involving a Cook, some Food, and a Heating Instrument, and is evoked by words such as bake, blanch, boil, broil, brown, simmer, steam, etc. 例如"加热(Apply heat)"框架描述一类常见场景：包含厨师、食物、加热器具；bake（烘烤）、blanch（焯烫）、boil（煮沸）、broil（炙烤）、brown（煎烤上色）、simmer（慢炖）、steam（蒸）等词均可唤起该框架。

We call these roles frame elements (FEs) and the frame-evoking words are LUs in the Apply heat frame. 我们把这些角色称为**框架元素(FE)**；唤起框架的词汇，就是该框架下的词汇单元。

Some frames are more abstract, such as Change position on a scale, which is evoked by LUs such as decline, decrease, gain, plummet, rise, etc., and has FEs such as Item, Attribute, Initial value and Final value. 部分框架更加抽象，例如"标度位置变化(Change position on a scale)"，decline（下降）、decrease（减少）、gain（增长）、plummet（暴跌）、rise（上升）等词汇单元可唤起该框架；框架元素包括项目、属性、初始值、终值。

In the simplest case, the frame-evoking LU is a verb and the FEs are its syntactic dependents: 最简单的情况：唤起框架的词汇单元是动词，框架元素就是它的句法从属成分：

[Cook Matilde] fried [Food the catfish] [Heating instrument in a heavy iron skillet]. [厨师 玛蒂尔德] 煎了 [食物 鲶鱼] [加热器具 在厚重的铁煎锅之中]。

[Item Colgate's stock] rose [Difference $3.64] [Final value to $49.94]. [项目 高露洁股票] 上涨了 [差值 3.64美元] [终值 至49.94美元]。

However, event nouns such as reduction in the Cause change of scalar position frame also evoke frames: or adjectives such as asleep in the Sleep frame: 但是事件名词（例如"标度致使变化"框架中的reduction"缩减"）同样可以唤起框架；形容词也可以，例如Sleep（睡眠）框架中的asleep（睡着的）：

...the reduction [Item of debt levels] [Value 2 to $665 million] [Value 1 from $2.6 billion] ……缩减 [项目 债务规模] [终值2 至6.65亿美元] [初值1 从26亿美元]

[Sleeper They] [Copula were] asleep [Duration for hours] [睡眠者 他们] [系动词 were] 睡着了 [时长 数小时]。

The lexical entry for a predicating word, derived from such annotations, identifies the frame which underlies a given meaning and specifies the ways in which FEs are realized in structures headed by the word. 由上述标注得到的谓词词条，会标明支撑该义项的底层框架，并且规定框架元素在该词中心的句法结构中如何实现。

Many common nouns, such as artifacts like hat or tower, typically serve as dependents rather than clearly evoking their own frames. 很多普通名词，例如帽子、塔楼这类人造物名词，大多充当从属成分，并不会清晰唤起自身专属框架。

The main purpose of annotating such items is to identify the most common predicates that govern phrases headed by them, and thus to illustrate the ways in which these common nouns function as FEs within frames evoked by the governing predicates. 标注这类名词，主要目的是识别支配该名词短语的高频谓词，以此展示普通名词如何作为框架元素，出现在支配谓词所唤起的框架内部。

We do recognize that artifact and natural kind nouns also have a minimal frame structure of their own. 我们承认：人造物、自然类名词本身也具备极简的框架结构。

For example, artifacts often occur together with expressions indicating their sub-type, the material of which they are made, their manner of production, and their purpose/use; these are defined as FEs in the frames for various types of artifacts. 例如人造物名词常搭配表示子类型、制作材料、生产方式、用途的表达；这些都被定义为各类人造物框架的框架元素。

Consider two example sentences from the Clothing frame. 请看Clothing（衣物）框架的两个例句。

(1) He took a packet of Woodbines out of the breast pocket of [his Wearer] [cotton Material] [shirt Garment] and lit one. (1) 他从【穿戴者 他的】【材料 棉质】【衣物 衬衫】胸前口袋掏出一包伍德拜恩香烟，点燃一支。

(2) She had a [white Descriptor] [silk Material] [blouse Garment] on, and a severe grey skirt that reached halfway down her calves. (2) 她穿着【描述词 白色】【材料 丝绸】【衣物 女式衬衫】，还有一条剪裁朴素的灰色半长裙，长度到小腿中部。

However, the frames evoked by artifact and natural kind nouns rarely dominate the clauses in which they occur, and so are seldom selected as targets of annotation. 但是人造物、自然类名词唤起的框架很少在小句中占据主导地位，因此很少被选为标注目标。

Formally, FrameNet annotations are constellations of triples that make up the frame element realization for each annotated sentence, each consisting of a frame element (for example, Food), a grammatical function (say, Object) and a phrase type (say, NP). 形式上，FrameNet标注由三元组集合构成，用来记录每个标注句的框架元素实现；三元组 = 框架元素（例如Food食物） + 语法功能（例如Object宾语） + 短语类型（例如NP名词短语）。

We think of these three types of annotation on each tagged frame element as layers and they are displayed as such in the annotation software used in the project. 我们把打标框架元素上这三类标注视为不同标注层；项目所用标注软件也按分层形式展示。

However, to avoid visual clutter, the grammatical function and phrase type layers are not displayed in the web-based report system. 但网页版报告系统不会展示语法功能层、短语类型层，避免界面信息繁杂。

The full data, available as part of the data download (see the FNdata link on the FrameNet homepage), include these three layers (and several more not discussed here) for all of the annotated sentences, along with complete frame and FE descriptions, frame-to-frame relations, and lexical entries summarizing the valence patterns for each annotated LU. 完整数据集可以下载（见FrameNet主页FNdata链接），包含全部标注句子的这三层信息（以及本文未讨论的其他多层标注），同时附带完整框架与框架元素描述、框架-框架关系、汇总每个词汇单元配价模式的词条。

FrameNet annotations derive from two sources. FrameNet标注有两大来源。

In pursuing the goal of recording the range of semantic and syntactic combinatory possibilities (valences) of each word in each of its senses, we normally concentrate on a particular target LU and extract sentences from the different texts of a corpus containing that LU. 为了记录词汇各个义项全部语义-句法组合可能性（配价），我们一般聚焦某一个目标词汇单元，从语料库各类文本中提取包含该词汇单元的句子。

Then we annotate a selection of the extracted sentences in respect to the target LU. 随后选取一部分提取出来的句子，针对该目标词汇单元做标注。

In another kind of work that represents a much smaller percentage of our overall annotations, we annotate running text. 另一类标注占比很小，即对连续全文文本做标注。

Full-text annotation differs from sentence annotation mostly in that the sentences are chosen for us, so to speak, by the author of the text. 全文标注与单句标注主要区别在于：句子是文本作者给定的，不是标注人员筛选出来的。

The annotation of running text is technically possible thanks to the annotation layering technique: FN lexicographers can one by one declare each word in a sentence a target, select a frame relative to which the new target is to be annotated, get a new set of annotation layers (frame element, grammatical function, phrase type) and appropriate frame element tags, and then annotate the relevant constituents. 依托分层标注技术，全文标注才得以实现：FrameNet词典编纂者可以把句子中每个词依次设为目标，选择用于该目标标注的对应框架，生成一套全新标注层（框架元素、语法功能、短语类型）与框架元素标签，再对相关成分完成标注。

1.1 Comparison with WordNet and ontologies

1.1 与WordNet及本体知识库对比

The FrameNet database is a lexical resource with unique characteristics that differentiate it from other resources such as commercially available dictionaries and thesauri as well as from the best-known on-line lexical resource, WordNet. FrameNet数据库是特色独特的词汇资源；与商用词典、类义词典，以及知名在线词汇资源WordNet均存在区别。

• Like dictionary subentries, FrameNet lexical units come with definitions, either from the Concise Oxford Dictionary, 10th Edition (courtesy of Oxford University Press) or a definition written by a FrameNet staff

member. • 和词典子条目类似，FrameNet词汇单元附带释义；释义来源为《简明牛津词典第10版》（牛津大学出版社授权），或是FrameNet工作人员撰写。

• Unlike commercial dictionaries, we provide multiple annotated examples of each sense of a word (i.e. each lexical unit). Moreover, the set of examples (approximately 20 per LU) illustrates all of the combinatorial possibilities of the lexical unit. • 和商用词典不同，我们为每个词义（即每个词汇单元）提供多条标注例句。每个词汇单元大约20条例句，覆盖该单元全部组合可能性。

The examples are attestations taken from naturalistic corpora, rather than constructed by a linguist or lexicographer. 所有例句均取自真实自然语料，而不是语言学家、词典编纂者人为编造。

The main FrameNet corpus is the 100-million-word British National Corpus (BNC, http://www.natcorp.ox.ac.uk), which is both large enough and balanced across genres (editorials, textbooks, advertisements, novels, sermons, etc.), but, of course, lacks many specifically American expressions. FrameNet主要语料库是一亿词规模的英国国家语料库BNC；语料规模大，语体均衡（社论、教材、广告、小说、布道文等），但缺少大量美式英语表达。

We also use U.S. newswire texts provided by the Linguistic Data Consortium http://www.ldc.upenn.edu, and the American National Corpus http://anc.org. 项目同时使用语言数据联盟LDC提供的美国新闻专线文本，以及美国国家语料库ANC。

• Our analysis of the English lexicon proceeds frame by frame rather than by lemma, whereas traditional dictionary-making proceeds word by word through the alphabet. • 我们对英语词汇的分析是以**框架为单位**推进，而不是以词元为单位；传统词典编纂是按照字母顺序逐词处理。

Thus, while a traditional lexicographer measures progress in words completed, FrameNet measures progress in frames completed. 因此传统词典编纂者以完成多少个词条衡量进度，FrameNet以完成多少个语义框架衡量进度。

Having one or more LUs for a given word in completed frames does not preclude the possibility that there could be other LUs for the same word in future frames. 某个词在已完成框架中拥有一个或多个词汇单元，不代表该词不会在后续新增框架中出现其他词汇单元。

Each lexical unit is linked to a semantic frame, and hence to the other words which evoke that frame. 每个词汇单元关联一个语义框架，进而与所有唤起同一框架的其他词汇关联。

This makes the FrameNet database similar to a thesaurus, grouping together semantically similar words. 这一点让FrameNet数据库和类义词典有相似之处，把语义相近的词汇聚合在一起。

WordNet and all ontologies provide some sort of hierarchical relations between their nodes; likewise, FrameNet includes a network of relations between frames. WordNet以及各类本体知识库在节点之间定义层级关系；FrameNet同样具备一套框架之间的关系网络。

Several types are defined, of which the most important are: 系统定义了多种关系，最重要的几类如下：

Inheritance: An IS-A relation. The child frame is a subtype of the parent frame, and each FE in the parent is bound to a corresponding FE in the child. 继承(Inheritance)：IS-A（是-一个）关系。子框架是父框架的子类型；父框架的每一个框架元素，都绑定到子框架对应的框架元素。

An example is the Revenge frame which inherits from the Rewards and punishments frame. 例子：复仇(Revenge)框架继承自奖惩(Rewards and punishments)框架。

– Using: The child frame presupposes the parent frame as background, e.g the Speed frame “uses” (or presupposes) the Motion frame; however, not all parent FEs need to be bound to child FEs. – 使用(Using)：子框架把父框架作为预设背景。例如速度(Speed)框架“使用”（预设）运动(Motion)框架；但父框架的框架元素不需要全部绑定到子框架。

Subframe: The child frame is a subevent of a complex event represented by the parent, e.g. the Criminal process frame has subframes of Arrest, Arraignment, Trial, and Sentencing. 子框架(Subframe)：子框架是父框架所代表复杂事件的子事件。例如刑事流程(Criminal process)框架包含逮捕、传讯、审判、量刑这些子框架。

Perspective on: The child frame provides a particular perspective on an un-perspectivized parent frame. 视角(Perspective on)：子框架针对无特定视角的父框架，提供某一个特定观察视角。

A pair of examples consists of the Hiring and Get a job frames, which perspectivize the Employment start frame from the Employer’s and the Employee’s point of view, respectively. 一组例子：雇佣(Hiring)与获得工作(Get a job)框架，分别从雇主、雇员视角，看待“开始雇佣(Employment start)”这个中性父框架。

These frame-to-frame relations are shown in the frame reports; the FE-to-FE relations are not shown in the frame reports but they can be viewed through the FrameGrapher tool (accessible from the FrameNet homepage) or at http://framenet.icsi.berkeley.edu/FrameGrapher. 框架报告中会展示框架-框架关系；框架元素之间的关系不会直接出现在框架报告，但可以通过FrameGrapher可视化工具查看，工具入口在FrameNet主页或直接访问网址。

Since we do not annotate many nouns denoting artifacts and natural kinds, the FrameNet database is not readily usable as an ontology of things. 由于我们没有大量标注人造物、自然类名词，FrameNet数据库并不适合直接充当事物本体知识库。

In this area, we mostly defer to WordNet, which provides extensive coverage, including hierarchical relations of areas such as animals, plants, etc. 这部分领域我们主要借助WordNet，它覆盖范围广，包含动物、植物等领域的层级关系。

1.2 What do we mean by word?

1.2 我们所说的“词”指的是什么？

In this discussion, we have used the word word in talking about lexical units. 前文讨论词汇单元时，我们一直使用“word（词）”这个说法。

The reality is actually rather complex. 但实际情况相当复杂。

When we say that the word bake is polysemous, we mean that the lemma bake.v (which has the word-forms bake, bakes, baked, and baking) is linked to three different frames: 当我们说bake是多义词，含义是：词元bake.v（词形包含bake、bakes、baked、baking）关联到三个不同框架：

• Apply heat: Michelle baked the potatoes for 45 minutes. • 加热框架：米歇尔烤土豆烤了45分钟。

• Cooking creation: Michelle baked her mother a cake for her birthday. • 烹饪造物框架：米歇尔为母亲烤了一个生日蛋糕。

• Absorb heat: The potatoes have to bake for more than 30 minutes. • 吸热框架：土豆需要受热烘烤超过30分钟。

These constitute three different LUs, with different definitions. 以上对应三个互不相同的词汇单元，各自拥有独立释义。

Multiword expressions such as given name and hyphenated words like shut-eye can also be LUs. 多词表达式（例如given name“名”）、带连字符词语（例如shut-eye“睡眠”）同样可以作为词汇单元。

Idiomatic phrases such as middle of nowhere and give the slip (to) are also defined as LUs in the appropriate frames (Isolated places and Evading, respectively), and their internal structure is not analyzed. 习语短语，如middle of nowhere（荒无人烟之地）、give the slip to（甩掉），同样被定义为对应框架（分别是偏僻地点、逃避框架）下的词汇单元，不对短语内部结构做分析。

For additional discussion also see the FAQs on the FrameNet website. 更多说明参见FrameNet网站常见问题FAQ。

Chapter 2 Frame Development

第2章 框架开发

The frames that make up the core of the project’s work do not come out of nowhere. 构成项目核心的语义框架并非凭空产生。

The project spends a considerable amount of time coming up with a description of each frame as a whole and each of its frame elements. 项目会投入大量时间，为每个框架整体、框架内每个框架元素撰写描述。

However, frames are models and groupings of ideas that are evoked by words, and those descriptions depend on decisions about the breadth of the vocabulary that we are modeling with a frame. 但框架只是模型，是词语唤起的概念集合；框架描述取决于我们决定该框架覆盖多大范围词汇。

Whether we start with an idea or start with a word, the core of the frame development process has always been looking at corpus attestations of a group of words that we believe to have some semantic overlap, and dividing these attestations into groups. 无论我们是从概念出发，还是从词语出发，框架开发的核心流程始终是：考察一组存在语义重叠词语的语料实例，再把这些语料实例划分成不同组别。

Afterward, we combine the small groups into large enough groupings to make reasonable frames at which point we may (equivalently) call the words targets, lexical units, or frame-evoking elements. 之后把小分组合并成规模合适的大分组，以此构建合理框架；此时分组内的词语可以等价称为标注目标、词汇单元、唤起框架的成分。

In the end we want to end up with groups of target words in each frame that have a particular kind of semantic overlap. 最终目标：每个框架内部的目标词集合，共享某一类特定语义重叠特征。

Speaking generally, all of the words should share a background of the kinds of questions they answer, the kinds of situations that are presupposed for sentences with the target, and the way that a speaker is thinking about a situation when they use the target. 总的来说，框架内全部词语应当共享一套背景：它们能够回答同类问题；使用该目标词的句子预设同类场景；说话人使用该词时，对场景的认知视角保持一致。

In the past, the criteria for such grouping have been informal and intuitive, but we have made an effort to make these criteria more explicit. 过去这类分组标准是非正式、依赖直觉的；现在我们努力把标准变得更加明确化。

In a practical sense, the criteria are of two kinds: 从实操层面看，标准分为两类：

1. A checklist of features, where if a criterion of similarity is not satisfied, we should put words in different frames, and
2. 一套特征检查清单：若不满足某条相似性标准，则词语应当归入不同框架；
3. A more difficult to define principle that our groupings should be useful, especially as paraphrases and as alternative answers to a question.
4. 另一条较难精确定义的原则：分组结果应当具备实用性，尤其要适合互相转述、作为同一问题的替换回答。

We discuss each of these in the following sections. 下面章节会逐一讨论。

2.1 Checklist for grouping words:

2.1 词语分组检查清单

The following checklist refers to features where all members of a frame should be identical or at least closely comparable. 下面这份清单描述特征：同一个框架下所有成员，在这些特征上应当完全一致，或至少高度可比。

In order of how easy they are to apply in practice and how categorical the similarity needs to be, they are: 按照实操难易程度、对相似性要求的严格程度排序：

1. Same FEs
2. 框架元素集合相同
3. Words denote same part of scene
4. 词语指向场景的同一组成部分
5. Same FE profiling
6. 对框架元素的侧显方式相同
7. Same relations
8. 关系集合相同
9. Similar types
10. 类型相似

We have endeavoured never to put two lexical units in the same frame if they do not have the same number and kind of frame elements. 我们坚持：如果两个词汇单元的框架元素数量、种类不一致，绝不放入同一个框架。

The criteria that fillers of a frame element should have a similar type for all LUs is much more difficult to judge: is the product of frying ontologically the same as the product of boiling? 但“同一个框架元素的填充成分类型应当相似”这条标准很难评判。例如：油炸得到的产物，和水煮得到的产物，本体论层面是否属于同一类？

They are clearly not identical, and we have made the judgement call that they are similar enough. 二者显然并不完全等同；但我们人工判定二者已经足够相似，可以放在同一框架。

2.1.1 Number and types of frame elements

2.1.1 框架元素的数量与类型

All LUs in a frame must have the same number and types of frame elements in both explicit and implicit (NI)
同一个框架下全部词汇单元，无论显式表达，还是隐式零实例化(NI)，框架元素的数量、类型必须统一。

2.1.1.1 Number

2.1.1.1 数量

If the number of essential, syntactically prominent frame elements differs from LU to LU or from sentence to sentence, this generally suggests that the frame should be split so that each resulting frame captures the difference. 如果核心、句法凸显的框架元素数量，随词汇单元、句子发生变化，一般就应当拆分框架，让拆分后的每个框架对应其中一类情况。

For instance, we have regularly split causatives from inchoatives. 典型实践：我们通常会把使成式和起始式拆分为不同框架。

Consider the verbs increase and diminish in examples (1) to (4) below. 请看下面例句(1)-(4)中的increase（上升）、diminish（下降）。

(1) Also, the violent crime rate has increased from 455 to 563 offenses per 100,000 population, despite a decline in crime rates nationally. (1) 此外，每十万人口暴力犯罪案件数从455起上升至563起，尽管全国犯罪率整体在下降。

(2) The law has increased the Government Employee Bonus calculation rate from 6% to 7%. (2) 该法案把政府雇员奖金核算比例从6%提升到7%。

(3) From the summer 1998 to the summer 1999 the speed variation has decreased. (3) 1998年夏至1999年夏，速度波动值出现下降。

(4) We have decreased the number of service calls. (4) 我们减少了服务呼叫的数量。

Examples (1) and (3) present the up or down changes in the value of a scalar attribute as events that happen by themselves, whereas examples (2) and (4) present them as caused by a Cause or an Agent. 例句(1)(3)把标度属性数值升降刻画为自发发生的事件；(2)(4)刻画为由某个诱因或者施事导致的事件。

We have separated the two types of sentences into two different frames, an inchoative frame called Change position on a scale for examples (1) and (3) and a causative one called Cause change of position on a scale for examples (2) and (4). 我们把两类句子分到两个框架；(1)(3)归属于起始式框架“标度位置变化”；(2)(4)归属于使成式框架“致使标度位置变化”。

There may be a legitimate objection about the presence of an Agent or Cause being just a vague linguistic intuition and that we ignore the fact that everything that happens is caused. 有人会合理反驳：区分施事/诱因只是模糊语言学直觉，世间发生的一切事件其实都存在诱因。

In this and in many other situations where one needs to decide between splitting and lumping causative and inchoative sentences, linguistic evidence suggests that splitting is the right way to proceed. 但在本案例，以及大量需要抉择是否拆分使成-起始句式的场景中，语言学证据都支持做拆分。

First, there will typically be lexical units that exhibit only one of the two uses. 第一，很多词汇单元只具备其中一类用法。

For instance, the verb gain only has inchoative uses when referring to scalar change, while the verb lower only allows causative uses in the domain of scalar change. 例如动词gain表示标度变化时只有起始式用法；而lower描述标度变化时，只允许使成式用法。

Second, cross-linguistic comparison also shows that other languages often distinguish inchoatives and causatives by derivational morphology. 第二，跨语言证据显示：其他语言经常依靠派生词形来区分起始式、使成式。

As pointed out above, we factor out language-wide constructions when figuring out how many participants there are in a particular use of a particular LU. 如上所述，在判断某个词汇单元某一用法的参与者数量时，我们会把语言通用构式的影响排除出去。

Thus, we don't need an extra frame for the verb sell to account for uses like Those boots sell well because the sentence involves subject selection construction (often called middle) which deprofiles and makes generic one or several of the prominent actors, in this case the Seller. 因此我们不需要为sell增设额外框架去处理"Those boots sell well（这靴子很好卖）"这类句子；该句用到主语选择构式（常称为中动构式），把其中凸显参与者（本句是卖方）取消侧显，泛化处理。

We then simply label Seller as CNI , deeming it still conceptually present, but generic. 我们只需将Seller（卖方）标记为CNI构式零实例化，认定该角色概念层面仍然存在，只是被泛化。

There is indeed linguistic evidence for the conceptual presence of the Seller: the notion of ease that is assessed requires a sentient being that could experience ease or difficulty selling. 的确有语言学证据证明卖方在概念层面存在：评价"好卖/难卖"，预设存在一个能够体会销售难易度的有感知主体。

Obviously, similar reasoning also applies to passives. 显然，这套推理逻辑同样适用于被动句。

In addition, we have used the idea that a construction can suppress the realization of a conceptually present frame element to account for the deleted objects in instructional imperatives, e.g. Place in oven for twenty minutes. 除此之外，我们借助"构式可以抑制概念上存在的框架元素的表层实现"这一思想，解释指令祈使句中宾语省略现象，例如"放入烤箱烤20分钟"。

2.1.1.2 Type

2.1.1.2 类型

The basic semantic type for a frame element ought to be broadly constant across uses. 同一个框架元素，在各类用法中基础语义类型应当大体保持一致。

If that is not so, it suggests the need to posit distinct frame elements. 如果不满足，则应当设立互相独立的框架元素。

In some cases, however, we still want to recognize a relationship between frame elements whose syntactic form suggests that they refer to ontologically different kinds of entities. 但部分情况下，句法形式显示两个框架元素指向本体论上不同实体，我们仍然希望识别二者之间存在关联。

For example, in I want [to win] compared with I want [an orange], both complements of the verb want have something to do with the Desiring frame, but each of the complements directly refers to something rather

different. 例如对比两句 I want [to win]（我想要赢）与 I want [an orange]（我想要一个橙子）；want的两个补语都和“欲望(Desiring)”框架相关，但补语直接指称的事物完全不同。

While the verb phrase to win refers to a type of event, the NP an orange refers to an entity. 动词短语to win指向一类事件；名词短语an orange指向一个实体。

Clearly, the NP-complements of want and other such predicates like it, are understood to metonymically stand for events centrally involving them: ‘to want an orange’ is typically ‘to want to eat an orange’. 很明显，want以及同类谓词的名词短语补语，会通过转喻，代表以该实体为核心的事件；“想要橙子”通常含义是“想要吃橙子”。

There is evidence that another event is to be understood by speaker and hearer when want is used with an object NP: sentences such as I want your report on my desk tomorrow morning include a time phrase that does not refer to the time of wanting but to the time when the desired event is to occur (or the desired state ought to hold). 有证据表明：当want接名词短语宾语时，听说双方都需要解读出另一个隐含事件。例句“I want your report on my desk tomorrow morning（我希望明天早上你的报告放到我桌上）”，句中时间短语修饰的不是“想要”这个心理活动，而是所期待事件/状态应当达成的时间。

When, as with want, some of the target’s complement types are appropriate to the ontological category of the frame element while others that are metonymically related are not, we normally create two distinct frame elements and relate them to each other via the Excludes relation (for which, see Section 3.2.2.3). 当目标谓词（如want）部分补语类型匹配框架元素本体类别，另一部分依靠转喻实现，我们通常创建两个独立框架元素，用Excludes互斥关系连接二者（参见3.2.2.3）。

Splitting in this manner is useful in several ways. 这样拆分有多方面好处。

It allows the right frame element to frame element relations because in some cases, related frames will allow only one of the (metonymically or otherwise related) frame elements. 可以保障框架元素之间关系设置正确；有些相关框架只允许使用其中某一个（转喻或非转喻）框架元素。

Having distinct frame elements also facilitates finding data to study Pustejovskian coercions. 互相独立的框架元素，也方便搜集语料研究普斯特若夫斯基提出的类型强制现象。

And finally, it makes the annotation task easier if the clear ontological distinction is maintained as a frame element distinction. 最后，把本体论上清晰的区分固化为框架元素区分，标注工作会变得更简单。

2.1.2 Aspectual Coherence

2.1.2 体连贯性

In aspectually complex frames, the lexical units should all entail the same set of stages and transitions. 在体结构复杂的框架内，全部词汇单元应当蕴含同一套事件阶段与状态转换。

For instance, while work on and develop both evoke the background of a sentient Agent expending effort to accomplish some Goal, only the latter entails (in past tense utterances) that the Goal was in fact accomplished. 举例：work on（致力于）和develop（开发）都唤起背景：有感知施事付出努力去达成某个目标；但只有develop在过去时态句子中蕴含“目标已经实际达成”。

Similarly, the verb shoot can be used to report an event of firing a projectile at a person and hitting, but it does not entail that the person dies in contrast to, for instance, the verb decapitate. 同理，shoot可以描述朝人发射投射物并且命中，但并不蕴含对方死亡；对比decapitate（斩首），后者必然蕴含死亡结果。

In all such cases, where lexical units differ in whether an intended and typical ‘fulfillment’ stage is in fact reached, we split frames. 只要词汇单元之间，在是否蕴含预期典型的完成阶段这件事上存在差异，就需要拆分框架。

As with the number and type of frame elements, the requirement that the same subevents are entailed for all lexical units also argues for separating causatives and inchoatives, since the former class involves an extra causing subevent that the latter class lacks. 和框架元素数量、类型的判定标准一样，“全部词汇单元蕴含相同子事件”这一要求，同样支持拆分使成式与起始式；使成式多了一个起始式不具备的致因子事件。

2.1.3 Perspective and Profiling

2.1.3 视角与侧显

The same frame elements will be profiled across all lexical units of a frame. 同一个框架内，所有词汇单元侧显的框架元素集合必须保持一致。

That is, the same participant’s point of view should be emphasized with all of them. 也就是说，框架内全部词语，所着重凸显的参与者视角必须统一。

Consider the difference between the verbs buy and sell: the first takes the Buyer’s perspective and the second the Seller’s perspective. 请看buy（购买）与sell（售卖）的区别：buy采用买方视角；sell采用卖方视角。

Note that when multiple perspectives on a type of scenario are possible, typically we relate the frames that carry particular perspectives to a non-perspectivized background/scenario frame via the Perspective on relation (see Sect. 6.1.2). 注意：当同一个场景可以具备多种视角，我们一般会把各个带特定视角的框架，通过Perspective-on视角关系，关联到一个不带特定视角的背景场景框架（参见6.1.2节）。

Thus, the perspectivized frames can be related to each other as candidates for paraphrasing via the background scenario. 因此各个带视角的框架，可以依托共同背景场景框架，建立互相转述的对应关系。

> Figure 2.1: Commercial Transaction 图2.1：商业交易关系图 （图内容：Reciprocality→Commercial_transaction；Commercial_transaction分别指向Commerce_goods-transfer与Commerce_money-transfer；Commerce_goods-transfer下分Commerce_buy、Commerce_sell；Commerce_money-transfer下分Commerce_collect、Commerce_pay）

2.1.4 Semantic Relationships

2.1.4 语义关系

2.1.4.1 Relations among frame elements

2.1.4.1 框架元素之间的关系

We want the interrelations between frame elements to be the same for all the LUs in a frame. 同一个框架内，所有词汇单元对应的框架元素之间相互关系必须保持一致。

Consider, for instance, the fact that the Purpose frame elements of buy and sell have a relationship to different participants: a Purpose expressed with buy relates to the Buyer, a Purpose expressed with sell relates to the Seller. 举例：buy和sell的Purpose（目的）框架元素，关联的参与者不一样；buy句中的目的关联买方；sell句中的目的关联卖方。

(5) I bought this thing [in order to test it out]. (5) 我买下这个东西【为了测试它】。

(6) At least you are selling them [in order to give the money to cancer research]. (6) 至少你在售卖它们【为了把所得款项捐给癌症研究】。

The difference in the orientation of the Purpose frame element parallels the difference in perspective that we noted above and supports the need for a frame distinction for buy and sell. 目的框架元素指向对象的差异，正好对应上面提到的视角差异；这也证明buy、sell应当分属不同框架。

2.1.4.2 Same relation to background frames

2.1.4.2 与背景框架之间的关系保持一致

The presuppositions, expectations, and concomitants of the targets within a frame will be shared. 同一个框架内部所有目标词，共享预设、预期、伴随事件。

For example, the verb cross-examine evokes a courtroom session, a preceding event of questioning by an opposing legal party, etc., differentiating it from the simpler examine. 例如动词cross-examine（交叉讯问）会唤起法庭庭审场景，预设对方法律方已经完成一轮讯问；这一点把它和简单的examine（讯问）区分开来。

By similar reasoning, the verbs receive and take would be differentiated, since receive presupposes another willing agent participating as the Donor and take does not. 同理receive（收到）和take（拿走）也应当区分；receive预设另一个自愿的施事充当赠予者；take不存在该预设。

2.1.5 Similarity of type

2.1.5 类型相似性

2.1.5.1 Targets

2.1.5.1 目标词

The basic denotation of the targets in a frame should be similar. 同一个框架中各个目标词的基础指称应当相似。

For this reason, adjectives such as blue and broken which refer to completely different kinds of states, are not in the same frame, even though they are distributionally very similar otherwise. 正因如此，blue（蓝色的）、broken（破损的）虽然句法分布高度相似，但二者描述完全不同类型状态，不能放在同一个框架。

Admittedly, this diagnostic is easy to apply in some cases and hard in others. 诚然，这条判定标准有些场景很好用，有些场景很难把握。

Note that it is not reliably possible to base frame distinctions mainly on selectional restrictions. 注意：不能主要依靠选择限制来划分框架，这并不可靠。

One may, for instance, conclude that two different frames are involved with blue and broken given that, in the naive physics sense, the kinds of entities that have color are much more numerous and spread out over a greater part of an ontology than the entities that can be broken (in the concrete physical sense). 举例：朴素物理意义上，可以拥有颜色的实体，在本体中范围远大于会发生物理破损的实体；凭这点可以把blue与broken分到不同框架。

However, within the Self-motion frame the verbs crawl, flit, slither, and walk have fairly different selectional restrictions, though they are grouped together in the same frame. 但在Self-motion（自身运动）框架内，crawl（爬行）、flit（掠过）、slither（蜿蜒滑行）、walk（行走）选择限制差异很大，但仍然归入同一个框架。

2.1.5.2 Frame elements

2.1.5.2 框架元素

The pre-specifications that the frame-evoking elements give to various frame elements will be similar. 唤起框架的词汇，对各个框架元素做出的前置限定条件应当具备相似性。

This heuristic was applied in the Mass-motion frame containing lexical units such as crowd.v, flock.v, pour.v, stream.v, swarm.v, troop.v, etc. 这条启发式标准被用于群体运动(Mass-motion)框架；该框架包含crowd（成群涌动）、flock（成群飞动）、pour（倾泻）、stream（流淌）、swarm（蜂拥）、troop（成群行进）等词汇单元。

These LUs are not simply treated as part of Self-motion since they all require that the moving entity be a mass theme, generally made up of many individuals. 这些词汇单元没有简单归入自身运动框架，因为它们都要求运动主体是由大量个体构成的集合性主题。

However, the desideratum that all the LUs in a frame share selectional restrictions on a frame element is often relaxed in practice, as it would result in very fine-grained frames that are impractical to manage. 但实践中经常放宽"框架内全部词汇单元对框架元素拥有相同选择限制"这一理想要求；如果严格执行，框架粒度会过细，不便于维护管理。

Consider, for instance, the requirement of the verb tie that the Connector be a relatively long, thin, flexible object. 举例子：动词tie（捆扎）要求Connector连接物是细长、柔韧物体。

Other verbs that refer to attaching one object to another (or, viewed symmetrically, both to each other) either make very different specifications (e.g. glue, staple) or none at all (e.g. attach). 其他表示"把物体互相固定"的动词，有的给出完全不同限定（glue胶合、staple钉合），有的完全不限定（attach附着）。

Users should be aware that frame distinctions made on this basis are quite subject to change, based on how useful we or our users have decided such distinctions are. 使用者应当意识到：基于选择限制做出的框架区分，后续很可能调整，取决于项目与用户评估该区分是否实用。

2.2 Criteria not used for frame splitting

2.2 不用于框架拆分的判定标准

Aside from factors that are used to group words into frames, there are those that never enter into consideration: 除了用于把词语归入框架的各类因素，下面这些因素**绝对不参与框架拆分决策**：

2.2.1 Grammar differences

2.2.1 语法差异

At all times, we lump together those groups whose semantic differences are due to general constructions of the language: 只要语义差异仅仅来自语言通用构式，我们就把它们合并到同一个框架：

• passive • 被动句

• middle • 中动句

• tense/aspect constructions • 时-体构式

• composition with extra-thematic frame elements • 和非题元框架元素组合

• mere part-of-speech differences • 单纯词性差异

The list does not include causativity, since, as argued above, it is lexically specific. 这份清单不包含使成性；前面已经论证，使成性属于词汇本身自带属性，不是通用构式。

2.2.2 Scalar or polar antonymy

2.2.2 标度极性反义关系

We group together words that are antonyms of each other. 互为反义词的词语，可以放在同一个框架。

For instance, the adjectives high and low are both in the Position on a scale frame. 例如形容词high（高）、low（低）同属"标度位置"框架。

Similarly, the verbs love and hate are both in the Experiencer subj frame. 同理动词love（爱）、hate（恨）同属"经验者作主语"框架。

However, so-called converses (Cruse 1986), relational opposites such as buy and sell which take opposite points of view on a single event, are placed into separate frames because they profile different (sets of) participants. 但是Cruse(1986)所说的交互反义词，例如buy与sell，针对同一个事件采用完全对立视角，需要分到不同框架，因为二者侧显的参与者集合不同。

Likewise, so-called reversive pairs such as tie and untie are placed in different frames since the members of the pair denote different kinds of actions basically. 同理，逆转反义词对tie（捆上）与untie（解开），二者描述本质不同动作，分属不同框架。

2.2.3 Usage differences

2.2.3 用法差异

Any differences purely in speech context. 仅仅来自言语语境的差异，不作为拆分依据：

• deixis (come vs. go) • 指示现象（come来 vs go去）

• register (botch vs. fuck up) • 语体差异（botch笨拙搞砸 vs fuck up搞砸，语体正式度不同）

• dialect (lorry vs. truck; fixture vs. regular season game) • 方言变体（lorry英式卡车 vs truck美式卡车）

• evaluation (criticize vs. praise; genius vs. moron) • 主观评价色彩（criticize批评 vs praise表扬）

We mark these kinds of differences with semantic types (see Sect. 6.2). 这类差异我们依靠**语义类型(Semantic Types)**标签标记，不拆分框架（参见6.2节）。

For example, the adjective generous in the Stinginess frame would be marked with the semantic type Positive judgment, while the adjective stingy would be assigned the semantic type Negative judgment. 举例：

Stinginess（吝啬）框架中，generous（慷慨）标记语义类型Positive judgment正面评价；stingy（吝啬）标记Negative judgment负面评价。

Note, however, that thus far we have not applied semantic types in all cases where this would be appropriate. 注意：到目前为止，并不是所有适合的场景都已经完整打上语义类型标签。

2.3 Paraphrasability and utility

2.3 可转述性与实用性

Finally, frame development focuses on paraphrasability (or near-paraphrasability) of words and multi-words. 最后，框架开发核心关注点之一：词语、多词表达之间的可转述（近义转述）能力。

That is to say, we ask whether one can more or less felicitously substitute one lexical unit for another and still evoke the same frame and express the same kinds of semantic roles as syntactic dependents of the new lexical unit. 也就是考察：一个词汇单元是否可以大致妥帖替换另一个；替换之后仍然唤起同一个框架，并且句法从属成分承担同类语义角色。

Frame development does not directly address the (near-)paraphrasability that may exist between whole utterances. 框架开发**不直接处理完整句子之间的（近义）转述关系**。

There are, for instance, many pairs of utterances where the total meaning of one member is built up compositionally from several frame evoking elements while the total meaning of the other derives just from a single lexical unit that combines within it the complex semantic structure that can be expressed with multiple frame evoking elements. 举例子：很多句子对，其中一句整体语义由多个唤起框架的成分组合得到；另一句仅仅依靠单个词汇单元，就把前者多个成分表达的复杂语义结构全部封装起来。

Clear cases of this situation include causative-inchoative pairs. 典型案例就是使成-起始句式对。

Sentence (7) is centered on the inchoative harden, but combines it with the extra-thematic frame element Cause, expressed by the phrase due to the hydration of the cement, which composes to a causative scenario. 句(7)以起始式动词harden（变硬）为中心，但叠加非题元框架元素Cause诱因（由短语due to the hydration of the cement“因为水泥水化”实现），组合得到使成性场景。

(7) The paste hardened due to hydration of the cement. (7) 膏体因为水泥水化而变硬。

(8) The hydration of the cement hardened the paste. (8) 水泥的水化作用让膏体变硬。

Note that although causative-inchoative pairs are not in the same frame, the FrameNet database provides an explicit link between the paired frames via a frame-to-frame relation Causative_of. 注意，尽管使成-起始句式分属不同框架；FrameNet数据库会通过Causative_of使成关系，把成对框架显式关联起来。

Similarly, sometimes the frame relation Using also connects frames whose targets can figure in utterances that are paraphrases of each other. 同理，Using使用关系也可以连接目标词能够构成转述句的两个框架。

Consider examples (9) and (10). 请看例句(9)(10)。

(9) I’ve communicated by email with US Airways on many occasions and have always gotten a reply within a few days. (9) 我多次通过邮件和全美航空沟通，总能几天之内收到回复。

(10) I’ve emailed US airways on many occasions and have always gotten a reply within a few days. (10) I’ve emailed US airways on many occasions and have always gotten a reply within a few days. (10) 我多次发邮件联

系全美航空，总能几天之内收到回复。

The lexical unit communicate in (9) belongs to the very general Communication frame, which is used or inherited by many frames. 例句(9)中的词汇单元 communicate 属于非常通用的“通信(Communication)”框架，很多框架会使用或者继承该框架。

The lexical unit e-mail in (10) belongs to the Contacting frame, which uses the Communication frame. 例句(10)的词汇单元 e-mail 属于“联系(Contacting)”框架，该框架使用了通信框架。

The two lexical units and frames are not equivalent, which can be illustrated easily with the following pair of past tense utterances: 这两个词汇单元以及所属框架并不等价，下面这组过去时句子就可以清晰说明：

(11) I e-mailed him my new phone number. (11) 我把我的新手机号发邮件发给了他。

(12) I communicated my new phone number to him by e-mail. (12) 我通过邮件把我的新手机号告知了他。

While (12) entails that the Recipient received the message, (11) does not carry the same entailment. 句(12)蕴含接收者收到了信息；句(11)却没有这层蕴含。

With Contacting, no actual successful communicative act is implied, only the successful completion of acts which could establish the communication. “联系”框架并不暗示通信行为实际成功，只代表完成了有可能建立通信的动作。

Another kind of paraphrase between utterances relies on world knowledge and inference. 另一类句子间转述依赖于世界知识与推理。

Consider the pair of utterances in (13)-(14). 请看(13)-(14)这组句子。

(13) The Denver tight end caught the ball in the end zone. (13) 丹佛队近端锋在达阵区接到球。

(14) Denver scored a touch down. (14) 丹佛队完成一次达阵得分。

Catching the ball in the end zone is a means of scoring a touch down, but catching and scoring have no conventional relation to each other outside of football, and even in the football context the two are not necessarily linked. 在达阵区接球是达阵得分的手段；但是接球与得分在橄榄球领域之外不存在约定俗成关联，就算在橄榄球场景，二者也并不必然绑定。

However, in some cases when inference is needed to establish that sentences are paraphrases of each other, the lexical material may still guide the inferential process to some extent. 不过，有些情况下句子之间的转述关系需要依靠推理，但词汇内容仍然可以在一定程度上引导推理过程。

Compare sentences (15), which denotes a means action, and (16), which denotes what is achieved by the means action. 对比句(15)（描述手段动作）和句(16)（描述该手段动作达成的结果）。

(15) You needed my help so I got on the bus. (15) 你需要我的帮助，于是我坐上公交车。

(16) You needed my help so I came. (16) 你需要我的帮助，于是我过来了。

Boarding a bus and arriving somewhere have no necessary relation to each other. 登上公交车和抵达某地二者之间不存在必然联系。

But the noun bus is a lexical unit in the Vehicle frame and that frame is linked to the Motion frame, which in turn is used by the Arriving frame that includes come. 但是名词 bus 是“交通工具(Vehicle)”框架的词汇单元；该

框架关联到“运动(Motion)”框架；而“到达(Arriving)”框架（包含 come）又使用运动框架。

The framal links between bus and come thus provide some evidence that a semantic equivalence may be intended. 因此 bus 和 come 之间的框架关联，提供了线索，提示此处或许意在表达语义等价。

2.4 Granularity of frame hierarchy

2.4 框架层级的粒度

Generally, FrameNet groups words only for paraphrasability between lexical units but not for paraphrasability of utterances. 总的来说，FrameNet只依据**词汇单元之间的可转述性**聚合词语，不依据完整句子的转述性。

There is, however, one kind of hybrid case where FrameNet groups lexical units in the same frame, even though the alternating lexical units do not participate in the same syntactic constructions and despite the difference in the relative prominence that the evoked frames have in the sentence. 但是存在一类混合特例：即便替换后的词汇单元不参与相同句法构式，并且二者在句中唤起的框架凸显程度不同，FrameNet依然会把它们归入同一框架。

Consider the following pair of sentences: 请看下面一组例句：

(17) She wisely sold the house. (17) 她明智地卖掉了这套房子。

(18) She was wise to sell the house. (18) 她卖掉这套房子，此举很明智。

The understandings of (17) and (18) are clearly similar. 句(17)与句(18)的解读显然很相近。

However, there is a difference between the sentences in terms of the relative salience of the two frames, Mental property and Commerce sell. 但是两个句子，在“心智属性(Mental property)”和“商品售卖(Commerce sell)”这两个框架的凸显度上存在差异。

(17) mainly reports an event of selling whereas (18) focuses on the assessment of what the selling behavior says about the seller’s mental abilities. 句(17)主要报道售卖事件；句(18)重点评价售卖行为所体现出卖方的心智能力。

The lie-test shows that this is so. “谎言测试法”可以证明这一点。

If a speaker challenges (17) by saying That’s a lie, the selling event is being contested, but if they challenge (18) by saying That’s a lie, what’s denied is the appropriateness of the assessment. 如果说话人用“That’s a lie（这是谎话）”质疑句(17)，被质疑的是售卖这件事本身；如果质疑句(18)，被否定的是该评价本身是否恰当。

This pragmatic difference fails to be captured by many current logical representation formalisms of adverb meanings, which suggest that the adverb has the clause in its scope. 很多当代刻画副词语义的逻辑形式化体系无法捕捉这一语用差异；这类体系认为副词的辖域覆盖整个小句。

While FrameNet’s practice of grouping the adjective and adverb in the same frame is in line with the logical-semantic similarity, it does not match the pragmatic-syntactic fact that the frame evoked by the syntactic head of the clause is more prominent. 虽然FrameNet将形容词与副词归入同一框架的做法契合逻辑-语义层面的相似性，但它和一条语用-句法事实并不匹配：小句句法中心所唤起的框架具备更高凸显度。

The ultimate test of these decisions is utility, and we believe that we largely achieve our goal of showing the basic semantic distinctions for the word senses we have described. 评判上述框架划分决策的最终标准是实用性；我们认为，我们大体上实现了目标，即展示所描写义项的基础语义差别。

It is perhaps obvious that many different FrameNets could be constructed which, arguably, follow the above principles as well as we do (or better), and yet diverge from FrameNet's set of frames in significant ways. 显而易见，完全可以构建出许多套不同的FrameNet，它们同样（甚至更好地）遵循以上全部原则，但最终得到的框架集合会与现有FrameNet存在显著差异。

For an example, see Kicktionary, a multilingual FrameNet specifically for soccer. 举个例子：Kicktionary，专门面向足球领域的多语言FrameNet资源。

This example makes clear that the most fundamental question is granularity. 该例子说明，最根本的问题就是**粒度**。

In many cases, we may have grouped words together that some set of users would prefer were separated. 很多场景下，我们把部分词语合并为一组，但有些用户会更希望将它们拆分开。

Semantic types differentiate some of these cases (see Negative). 语义类型标签可以对部分这类情况做区分（参见Negative相关章节）。

However, there are many other distinctions that users might want: different frames for eating and drinking, different frames for friends and spouses, etc. 但还存在大量用户期待的区分：例如把吃、喝分属不同框架，朋友与配偶分属不同框架等等。

In such cases, the project has made an assessment that the amount of work and or complexity introduced by these finer-grained distinctions is not worth the loss of similarity provided by putting them in the same frame. 遇到这类情况，项目会做评估：如果采用过细的区分，会带来巨大工作量与复杂度，而从同一框架内可以获得的语义相似性收益却会损失，因此不值得这么做。

(In principle, we could have our cake and eat it too by making very fine-grained frames for each word or smaller group of words, then making a generalizing frame similar to the current one which these smaller frames inherit from. This is considerable additional work, especially when many users seem to be looking for a categorical distinction of same vs. different.) （原则上我们可以两全其美：为每个词或者小规模词群建立细粒度子框架，再构建一个类似现有版本的概括性父框架，子框架全部继承自父框架。但这会带来巨大额外工作量，尤其多数用户仅仅需要简单的"相同/不同"二元区分。）

In the direction of coarser granularity, there are also users who wonder why we have made more than one frame in the Motion domain. 而朝着更粗粒度的方向，也有部分用户会疑惑：为什么运动相关领域要设置多个不同框架。

(Ruppenhofer et al. (2010) reports on software designed to automatically collapse fine distinctions in FrameNet to produce coarser granularity.) （Ruppenhofer等人2010年论文介绍了一款软件，可以自动合并FrameNet内部细粒度区分，输出粗粒度版本。）

The ultimate test of these decisions is utility, and we believe that we largely achieve our goal of showing the basic semantic distinctions for the word senses we have described. 评判这些决策的最终标准是实用性；我们认为我们大体实现了目标，即展现所处理各义项的基础语义差异。

The FrameNet Project is quite open to concrete suggestions from the community for changing the granularity of frames, in the likely event that better or more useful groupings of words can be discovered. FrameNet项目欢迎社区给出具体建议来调整框架粒度，前提是能够找到更好、更实用的词语分组方案。

Chapter 3 FrameNet Annotation

第3章 FrameNet标注

3.1 Introduction

3.1 简介

As a technical matter, the way in which FrameNet analyzes instances of a target predicate consists of marking up parallel aligned layers of annotation with appropriate label sets, as shown in Figure 3.1. 从技术实现角度看，FrameNet分析目标谓词实例的方式，是使用配套标签集，对多层相互对齐的平行标注层做标记，见图3.1。

The layers that are displayed in the FN Desktop can be manually selected by annotators. 标注人员可以在FrameNet桌面客户端手动选择需要显示的标注层。

The number of layers and the kind of information that can be recorded on them is technically unlimited. 标注层的数量、每层可记录的信息类型，在技术层面没有限制。

However, the four most important annotation layers are the Target, frame element (FE), grammatical function (GF), and phrase type (PT) layers. 但最重要的四层标注分别是：目标层、框架元素（FE）层、语法功能（GF）层、短语类型（PT）层。

On the first, the (parts of the) target predicate are marked while on the latter three, labels are applied to the constituents expressing the frame elements of the target. 目标层标记目标谓词（或谓词片段）；后三层用来给表达目标框架元素的句法成分打上标签。

The next-most important set of layers consists of the layers called Other; a layer called either Noun, Verb, Adj, Adv, or Prep depending on the part of speech of the target (this layer is also often called the part-of-speech-specific layer); and the Sent(sentence) layer. 次重要的一组标注层包括：Other层；根据目标词性分别命名的Noun / Verb / Adj / Adv / Prep层（常被称为词性专属层）；Sent（句子）层。

The Other layer holds labels relating to certain special constructional contexts in which the target may occur, such as relative clauses, existential constructions, and extraposition constructions. Other层存放特殊构式上下文相关标签，例如关系小句、存现构式、外置构式。

The part-of-speech-specific layer holds labels that can occur only with predicates of a particular part of speech. For instance, the Copula does not occur with verbal targets; it can be applied on the Noun, Adj, and Prep layers of targets of the appropriate part of speech. 词性专属层存放仅某一类词性谓词才会使用的标签。例如系动词Copula不会出现在动词目标层；仅可以在名词、形容词、介词目标对应的专属层使用。

The Sent(sentence) layer is special in that it does not bear any annotation labels: when the layer is invoked, information about the sentence as a whole can be recorded on the appearing list of check-boxes. Sent（句子）层比较特殊，它本身不带标注标签；启用该层之后，可以在弹出的复选框列表中记录整句相关信息。

A final group of layers includes, among others, layers holding labels related to part of speech (POS) and Named Entity Recognition (NER). 最后一类标注层包含词性POS层、命名实体识别NER层等。

This information is derived automatically from our corpora and third-party software and is generally not modified by FrameNet annotators. 这类信息由语料库与第三方软件自动生成，FrameNet标注人员一般不对其做修改。

We now turn to the details of the annotation process. 下面我们介绍标注流程细节。

The work can be divided into two kinds according to the way in which sentences are chosen for annotation. 按照句子的选取方式，标注工作分为两大类。

In the lexicographic annotation mode, our main focus is on the goal of recording the range of semantic and syntactic combinatory possibilities (valences) of each word in each of its senses. 词典编纂式标注模式：核心目标是记录每个词语各个义项全部的语义-句法组合可能性，也就是配价。

To that end, we extract sentences from the different texts of a corpus containing that LU. 为此，我们从语料库各类文本中提取包含该词汇单元的句子。

Then we annotate a selection of the extracted sentences in respect to the target LU. 之后挑选部分提取出来的句子，针对目标词汇单元开展标注。

In another kind of work that represents a much smaller percentage of our overall annotations, we annotate running text. 另一类占比很小的工作，即连续全文文本标注。

Full-text annotation differs from sentence annotation mostly in that the sentences are chosen for us, so to speak, by the author of the text. 全文标注和单句标注最大区别在于：句子由文本作者给定，而非标注人员筛选。

The annotation of running text is technically possible thanks to the annotation layering technique: FN lexicographers can one by one declare each word in a sentence a target, select a frame relative to which the new target is to be annotated, get a new set of annotation layers (frame element, grammatical function, phrase type) and appropriate frame element tags, and then annotate the relevant constituents. 依托分层标注技术，全文标注才得以实现：FrameNet词典编纂者可以依次将句中每个词设置为标注目标，选择用于该目标的语义框架，生成全新一套FE、GF、PT标注层与框架元素标签，再对相关句法成分完成标注。

> Figure 3.1: Annotation layers 图3.1：标注层示意图

Before going further into the details of annotation, let us briefly consider the Revenge frame, which will figure as an example frame repeatedly in this chapter. 在深入标注细节之前，我们简单看一下复仇(Revenge)框架；本章会反复拿它作为示例框架。

The definition of this frame follows: 该框架定义如下：

An Avenger performs some Punishment on an Offender as a response to an earlier action, the Injury, that was inflicted on an Injured party. 复仇者对冒犯者施加某种惩罚，以此回应此前发生在受害方身上的伤害事件。

The Avenger need not be identical to the Injured party but needs to consider the prior action of the Offender a wrong. 复仇者不一定就是受害方本人，但复仇者必须认为冒犯者之前的行为属于过错。

Importantly, the punishment of the Offender by the Avenger is seen as justified by individual or group opinion rather than by law. 重要一点：复仇者对冒犯者施加的惩罚，是依据个人或群体观念判定为正当，而不是依据法律。

Thus, the frame elements in the Revenge frame are Avenger, Punishment, Offender, Injury, and Injured party. 因此复仇框架的框架元素有：复仇者、惩罚、冒犯者、伤害事件、受害方。

Lexical units in this frame include avenge.v, avenger.n, get even.v, retaliate.v, retaliation.n, retribution.n, retributive.a, retributory.a, revenge.v, revenge.n, revengeful.a, revenger.n, vengeance.n, vengeful.a, and vindictive.a. 该框架下词汇单元包括：avenge.v、avenger.n、get even.v、retaliate.v、retaliation.n、

retribution.n、retributive.a、retributory.a、revenge.v、revenge.n、revengeful.a、revenger.n、vengeance.n、vengeful.a、vindictive.a。

Some example sentences with the lexical unit avenge are given here. 下面给出avenge词汇单元的部分例句。

(1) [His brothers Avenger] avenged [him Injured party]. (1) [复仇者 他的兄弟们] 为 [受害方 他] 报了仇。

(2) With this, [El Cid Agent] at once avenged [the death of his son Injury]. (2) 借此机会，[施事 埃尔·西德] 立刻为 [伤害事件 儿子之死] 复仇。

(3) [Hook Avenger] tries to avenge [himself Injured party] [on Peter Pan Offender] [by becoming a second and better father Punishment]. (3) [复仇者 胡克] 试图向 [冒犯者 彼得·潘] 为 [受害方 他自己] 复仇，[惩罚方式 成为一个更优秀的第二位父亲]。

Regardless of the kind of annotation that is done, the following guidelines apply to the annotation relative to a particular instance of a target word. 无论采用哪一类标注模式，针对目标词某一个实例做标注，都要遵守下面的准则。

FrameNet annotation is directed toward the dependents of a target word. FrameNet标注针对目标词的从属成分开展。

We do not annotate constituents that are understood only through context as referring to the filler of a particular frame element role. 如果某个成分只能依靠上下文语境才能解读为某个框架元素的填充项，则不对该成分标注。

Consider the following example. 请看下面例句。

(4) Smith was surprised when Lowry retaliated for the attack. (4) 当劳里为这次袭击展开报复时，史密斯感到很吃惊。

The target retaliate evokes the Revenge frame introduced above, but the surface syntax does not contain constituents representing the Offender or the Injured party. 目标词retaliate唤起上面介绍的复仇框架，但表层句法中不存在代表冒犯者、受害方的句法成分。

One plausible understanding of the sentence as a whole is that Smith was the one against whom Lowry retaliated. 对整句一种合理解读是：劳里报复的对象就是史密斯。

One might, thus, want to tag Smith as the Offender. 于是有人会想要把Smith标记为Offender冒犯者。

However, there is an alternative interpretation: Smith could just be an observer of a conflict involving Lowry and a third party. 但是还存在另一种解读：史密斯仅仅只是旁观者，冲突发生在劳里和第三方之间。

In such cases, where there is no syntactic construction present that guarantees that interpretation, we do not annotate the non-local phrases in question. 遇到这类没有句法构式来锁定该解读的情况，我们不会对这类非局域短语做标注。

We mark the Offender as omitted under definite null instantiation (DNI). 我们将Offender标记为确定式零实例化（DNI），表示该角色被省略。

We annotate whole constituents rather than just the head words of the target's syntactic dependents. 我们标注完整句法成分，而不是仅仅标注目标从属成分的中心词。

For instance, in example (3) above, the preposition on is included in the constituent expressing the Offender frame element, on Peter Pan. 例如上面例句(3)，介词on也被包含在表达Offender框架元素的成分on Peter Pan之内。

• Each dependent is annotated for Frame Element identity, Phrase Type, and Grammatical Function relative to the target LU. • 每一个从属成分，都要针对目标词汇单元标注：框架元素身份、短语类型、语法功能。

While the above guidelines apply equally to lexicographic and full-text annotation, there are also some clear differences between the two modes of annotation. 以上准则同时适用于词典编纂标注和全文标注；但两种模式之间同样存在明确差异。

• In FrameNet's lexicographic work, annotation is done relative to only one lexical unit per sentence, the target. (As noted previously, the target is typically a single word but can also be a multi-word expression such as a phrasal verb (e.g., give in) or an idiom (e.g., kick the bucket).) • 在词典编纂式标注中，一个句子只针对一个词汇单元（即标注目标）进行标注。（前文提到，目标一般是单个词，但也可以是多词表达，例如短语动词give in，习语kick the bucket。）

• In full-text annotation, by contrast, all contentful lexical units are treated as targets and their dependents are annotated. • 与之相反，全文标注中，所有实义词汇单元都会被当作目标，它们的从属成分都会被标注。

• In full-text annotation, we have no choice but to annotate each instance of a target word whether or not it contributes useful lexicographic information about the target. • 在全文标注中，无论目标词实例是否能够产出有价值的词典学信息，都必须做标注。

For instance, in a given text the LU may occur in complicated syntactic constructions that add nothing to our understanding of the target's meaning. 例如文本中某个词汇单元出现在复杂句法构式中，但该构式对理解词义没有增益。

A particular use of the LU may be very poor style or borderline ungrammatical. 词汇单元的某一处用法可能语体极差，或者近乎不合语法。

Some instances only provide more exemplification of an already amply documented argument realization pattern for the LU rather than exhibiting a rarer pattern. 有些实例仅仅是重复展示已经充分记录的论元实现模式，并没有展示罕见模式。

This contrasts with our sentence-annotation (lexicographic) work, where it is intentionally part of our annotation principles to avoid such sentences. 这和单句词典编纂标注形成对比：词典标注原则会刻意避开这类句子。

• In the lexicographic work, we intend the set of sentences annotated for a given LU to represent the full range of combinatorial possibilities for that LU. • 词典编纂标注模式下，为某个词汇单元挑选的全部标注例句，目标是覆盖该词汇单元全部组合可能性。

A given text, even of considerable length, is not likely to exemplify all patterns in which a target may occur. 单篇文本哪怕篇幅很长，也很难覆盖目标词所有的使用模式。

• The annotation contained in the FrameNet database does not offer information about frequency of occurrence. • FrameNet数据库内部的标注数据，不提供词语出现频次信息。

Full-text annotation necessarily means that we annotate all instances of any given contentful target. Thus, at least relative to a particular text, frequency information for LUs can be derived. 全文标注则必须标注实义目标

词的全部实例。因此至少针对某一篇特定文本，可以从中统计得到词汇单元频次信息。

So far, we have sub-divided our work into two parts depending on whether we choose the sentences to annotate or whether they are chosen for us by a text. 到这里，我们已经按照"人工挑选句子"还是"文本自带句子"把工作分为两大类。

Another subdivision applies specifically to our lexicographic work where we produce annotation of two different types, reflecting two different kinds of target words: 词典编纂标注内部还可以进一步划分，产出两类标注，对应两类不同目标词：

annotation relative to a frame bearing syntactic governor, either a predicate, modifier, or a referring expression and 一类：针对承载框架的句法支配成分做标注，可以是谓词、修饰成分、指称表达式；

annotation relative to a slot-filler, that is, relative to a referring expression that is a frame element of a frame determined not by itself but by a governor 二类：针对槽填充项做标注；即某个指称表达式，它本身不是唤起框架的成分，而是充当由支配成分所确定框架的框架元素。

Both kinds of annotation will be discussed. Since the annotation relative to frame-bearing syntactic governors is FrameNet's main task, it will be discussed first and in greater detail in Sections 3.2-3.7. Annotation relative to slot-fillers is discussed in Section 3.9. 两类标注都会展开讨论。针对承载框架的支配成分是FrameNet主要任务，将在3.2-3.7节优先详细说明；槽填充项标注放在3.9节。

3.2 Lexicographically motivated annotation practices

3.2 词典编纂导向的标注规范

In accord with FrameNet's goals, syntactic and semantic descriptions are tailored to lexicographic description only and may differ from regular linguistic analysis as well as from shallow semantic analysis in several ways. 按照FrameNet项目目标，句法、语义描写仅服务于词典编纂，会在多个方面和常规语言学分析、浅层语义分析存在区别。

• Typically, the analysis of sentences is only partial in the sense that we apply labels just to some words or phrases of a sentence. Although many sentences contain multiple frames evoking lexical units, in normal annotation we annotate only with respect to one of them. • 典型特点：句子分析是局部分析，只给句中部分词、短语打标签。虽然很多句子包含多个唤起框架的词汇单元，常规标注仅针对其中一个开展。

• We tag whole constituents that realize frame elements relative to our target words, rather than just tagging the head words of these constituents. That is, we work with a phrase structure grammar, rather than a dependency grammar. • 我们标记实现框架元素的完整句法成分，而不是只标记成分中心词。也就是说，我们使用短语结构语法体系，而非依存语法。

2As will be discussed in Section 3.2, we also sometimes apply frame element labels on a secondary frame element annotation layer that are not paired with Phrase Type and Grammatical Function labels. These annotations do not contradict the statement concerning the annotation of dependents with triples of FE, PT, and GF since they involve words or phrases that are not themselves direct dependents of the target. [2]正如3.2节会讲到，我们有时会在第二层框架元素标注层打FE标签，这类标签不搭配短语类型、语法功能标签。这类标注并不违背"从属成分使用FE-PT-GF三元组标注"这一规则，因为这些词语、短语本身并不是目标词的直接从属成分。

A consequence of this is that many frame element labels cover words that have no direct relation of their own to the target, but only to the head of their constituent. 这带来一个结果：很多框架元素标签会覆盖一些和目标词没有直接关系的词语，这些词语仅仅和所在成分的中心词有关系。

For instance, when a frame element is expressed by a noun which takes adjectival, prepositional or clausal complements or which is modified by such elements, these complements and modifiers are included in the frame element tag. 举例，如果框架元素由名词实现，该名词带有形容词、介词短语、小句补语，或是被这些成分修饰；那么这些补语、修饰成分全部被纳入框架元素标签覆盖范围。

Consequently, phrase type (and grammatical function) labels always cover full constituents. 因此短语类型标签、语法功能标签永远覆盖完整句法成分。

In (5), we tag both the noun story and the whole NP it heads; in (6), we tag both the noun fact and the full NP it heads; in (7), the with-PP is part of the frame element that is tagged. 句(5)中，名词story以及它所统领的完整名词短语都会被标记；句(6)中名词fact及其完整NP都会标记；句(7)中with引导的介词短语也属于被标记框架元素的一部分。

(5) I heard [a story about a man named Jed NP]. (5) 我听到了[一个关于名叫杰德的男人 NP]的故事。

(6) [The fact that I wear a funny shirt Sfin] is totally irrelevant. (6) [我穿着一件滑稽衬衫这件事 Sfin]完全无关紧要。

(7) I have [a cat with orange stripes NP]. (7) 我养了[一只带有橘色条纹的猫 NP]。

FrameNet syntax also differs from traditional treatments of subcategorization in paying attention to relational modifiers, including the non-heads of compounds. FrameNet句法描写和传统次范畴化描写另一个不同点：它关注关系型修饰成分，也包括复合词内部的非中心成分。

For example, the FrameNet description of a noun like clinic makes reference to the types of nouns which can modify this noun in compounds like allergy clinic. 举例，对于clinic（诊所）这类名词，FrameNet描写会记录可以在复合词（例如allergy clinic过敏诊所）中修饰它的名词类型。

In theoretical treatments, modifiers occurring with nouns would typically be outside the realm of subcategorization. 在理论语言学分析中，名词的修饰成分一般不属于次范畴化的研究范畴。

Thus, you can treat an allergy, treat the flu, and so forth. Modifiers occurring with the noun treatment often express the same role; thus, there are allergy treatments, flu treatments, etc. 动词treat可以搭配宾语allergy、flu等等。名词treatment的修饰成分经常表达完全相同的语义角色；于是就有allergy treatments（过敏治疗）、flu treatments（流感治疗）这类表达。

Such observations are of lexicographic relevance, and our syntactic descriptions reflect this perspective. 这类观察具备词典编纂价值，因此我们的句法描写会体现该视角。

In some situations, there are differences between the syntactic and semantic headedness of a clause. 部分场景下，小句的句法中心和语义中心并不一致。

For instance, Pa gave her a lecture has give as its syntactic head. However, from a semantic point of view the sentence reports an act of lecturing, not one of giving. 例如句子Pa gave her a lecture，句法中心是give；但从语义角度，句子描述的是“讲课”事件，而不是“给予”事件。

We understand cases such as give a lecture as involving frame-evoking nouns that are syntactically "supported" by verbs in order to project clauses. 我们把give a lecture这类情况理解为：唤起框架的名词，依靠动词做句法"支撑"，从而投射出完整小句。

This analysis is strengthened by the fact that noun+support verb combinations typically involve the selection of the support verb by the frame-evoking element rather than the other way around. 名词+支撑动词组合，是唤起框架的名词来挑选支撑动词，而不是动词挑选名词；该事实进一步支撑上面这套分析。

An important consequence of this analysis is the annotation of subjects of support verbs as frame elements relative to the noun. 这套分析带来一条重要推论：支撑动词的主语，要相对于名词本身标注为框架元素。

Thus, the sentence Pa gave her a lecture would be annotated with respect to the target lecture, with gave marked as a support verb, Pa marked as the Speaker and her marked as the Addressee. 因此句子Pa gave her a lecture要以lecture为标注目标；gave标记为支撑动词，Pa标记为说话人，her标记为听话人。

(See also sec. 3.2.7.) （参见3.2.7节。）

• Syntacticians commonly divide the complements of a predicate into arguments and adjuncts, or arguments and modifiers. In FrameNet, the semantic spirit of this distinction is covered by the distinctions of coreness status rather than through GFs (cf. Section 3.2.1). • 句法学家通常会把谓词补语分为论元与附加语（论元与修饰语）。在FrameNet中，该区分的语义内核由**核心性状态(coreness status)**来承载，而不是依靠语法功能标签（参见3.2.1节）。

As a consequence, we only have a grammatical function Dep (dependent), rather than distinguishing between the two types of complements. 因此我们只设置一个语法功能Dep（dependent从属成分），不对两类补语做语法功能层面区分。

• In general, we select sentences for annotation where, with the exception of subjects, all frame elements are realized locally by constituents that are part of the maximal phrase headed by the target word. • 总体上我们挑选标注例句遵循：除主语之外，全部框架元素都由目标词所统领最大短语内部的局域成分实现。

In addition, there are several clear cases, typically involving subjects, where the combination of particular lexical items and grammatical structures guarantees a particular interpretation of phrases which are not dominated by the maximal phrase headed by the target. 但也存在几类明确例外，大多和主语相关：特定词汇项搭配特定句法结构，可以让不受目标最大短语统制的短语获得确定解读。

One such example is that of modal verbs, which behave as so-called raising verbs. 典型例子就是情态动词，属于所谓的提升动词。

The subjects of modals such as may, will etc. always need to be interpreted as subjects and frame elements of the predicates that follow the modal. may、will这类情态动词的主语，解读上必须作为情态动词后面谓词的主语与框架元素。

For an example like [Bill SELLER] may sell it at that price, a treatment motivated by syntactic theory might suggest that the subject slot of sell is filled by a kind of trace, or that one of its required arguments is passed on to the higher raising predicate via a valence passing principle. 拿句子[Bill SELLER] may sell it at that price举例，句法理论分析会认为sell的主语位置由语迹填充，或者sell的论元通过配价传递原则向上传递给高层提升谓词may。

Under such a treatment, it could be left up to grammatical principles to establish the connection between the valence requirement of the target word and the fillers of those roles. 按照这套分析，目标词配价需求与角色填

充项之间的联系交给句法原则处理即可。

From a lexicographic point of view it is more useful to directly report what the semantic properties of the fillers of particular Frame Elements are, rather than to ask users to employ additional tools to establish lexical antecedents or valence fillers. 但从词典编纂视角看，直接记录框架元素填充项的语义属性更加实用，而不是让使用者额外使用工具去还原论元先行词、论元填充项。

• For most targets, frame elements are marked only on a single frame element layer. Under two kinds of circumstances, however, we record frame elements on an additional, secondary frame element annotation layer. • 绝大多数目标词，框架元素只标记在第一层FE层。但存在两种情况，我们会使用额外的第二层FE标注层记录框架元素。

3The first type of situation for which second-layer annotation is used is frame element conflation(Sec. 3.2.5), when a phrase that directly expresses one frame element also allows inferences about a second frame element. [3]第一种场景：框架元素融合（见3.2.5节），某个句法成分直接表达一个框架元素，同时还可以推导出第二个框架元素的信息。

Consider as an example the sentence Doctor Miller managed to at least cure [the epileptic]. 举例子：Doctor Miller managed to at least cure [the epileptic]（米勒医生至少治愈了这名癫痫患者）。

Out of context, this sentence is most naturally understood to mean that the epileptic is cured of epilepsy. 脱离上下文，这句话最自然解读是：这名癫痫患者的癫痫病症被治愈。

We tag the phrase the epileptic as Patient on the first frame element annotation layer and as Affliction on the second annotation layer. 于是我们把the epileptic在第一层标记为Patient（病人），第二层标记为Affliction（病症）。

Notice that when a core frame element, such as Affliction in the example of cure, is applied on a secondary annotation layer (cf. Sec. 3.2.5), no null instantiation tag is used concurrently. 注意，如果像本句Affliction这类核心框架元素标记在第二层标注层，就不会同时使用零实例化标签。

Given an appropriate context, all inferences in conflation contexts can be defeated. In cases where the inference is defeated, no secondary layer tagging is applied. 给定合适上下文，融合场景下的推导可以被取消；一旦推导被取消，则不再打第二层标签。

In the modified sentence Both the epileptic and the schizophrenic patient had terrible head colds last week and Doctor Miller managed to at least cure [the epileptic], the Affliction frame element is marked as null-instantiated rather than applied to the constituent the epileptic on a second frame element layer. 改写后的句子：Both the epileptic and the schizophrenic patient had terrible head colds last week and Doctor Miller managed to at least cure [the epileptic]。此时Affliction病症不再打第二层标签，而是标记为零实例化。

Note that frame element conflation is the only context where FN annotation marks phrases that are only inferred, rather than grammatically guaranteed, to express information about frame elements. 注意：框架元素融合，是FrameNet唯一一种给仅靠推导（而非句法强制保证）得到的框架元素信息打标的场景。

A second type of circumstance where FN annotation makes use of a secondary frame element layer has to do with possessor phrases that are obligatorily co-referential with another argument. 第二种使用第二层标注层的场景：领属短语必须和另一个论元共指。

Most commonly this arises with body part possessors such as in the Experience bodily harm frame. 该情况最常出现在身体伤害(Experience bodily harm)框架中的身体部位领属成分。

With verbs such as stub, sprain or strain the referent of the Experiencer frame element has to be expressed again as a possessive determiner in the Body part frame element and we tag the possessive on the second frame element layer with another instance of the frame element Experiencer. 对于stub（磕碰到）、sprain（扭伤）、strain（拉伤）这类动词，Experiencer经验者的指称，会再次以领属限定词形式出现在Body part身体部位框架元素中；此时把这个领属词在第二层标记为Experiencer。

Thus, in Ike cut [his finger] on the envelope, the subject NP Ike is marked as Experiencer on the first annotation layer, the phrase his finger is marked as Body part, and the possessive his is marked as Experiencer on the second frame element layer. 例如句子Ike cut [his finger] on the envelope，主语Ike第一层标记Experiencer；his finger标记为Body part；领属词his在第二层标记Experiencer。

In all cases, a label applied on the second layer is always co-extensive with, or contained in the boundaries of a label on the first FE layer; no word or constituent can be covered by a frame element label on the second layer alone. 所有情况下，第二层标签的覆盖范围，要么和第一层FE标签重合，要么完全包含在第一层标签边界之内；不存在只被第二层标签覆盖的词语或成分。

Finally, secondary frame element layers are never accompanied by grammatical function or phrase type layers; GF and PT layers are available only for first layer annotation. 最后，第二层FE层永远不搭配语法功能、短语类型标签；GF、PT只用于第一层标注。

The combination of our surface-oriented annotation and our commitment to providing grammatical function and phrase type information about the phrases that fill frame element roles causes some problems in the case of discontinuous frame elements. 我们采用面向表层的标注，同时要为框架元素填充短语提供GF、PT信息；遇到非连续框架元素时会带来一些问题。

For instance, with wh-extraction out of a frame element such as [Who] did you talk [about]?, we need to provide a grammatical function tag for the fronted wh-word who even though it does not bear a grammatical relation to the target word talk. 例如特殊疑问词移出框架元素的句子[Who] did you talk [about]?，前置的who和目标词talk没有直接语法关系，但依然要为who分配语法功能标签。

This and other cases of discontinuous frame element realization are discussed below in Section 3.2.8. 该案例以及其他非连续框架元素实现案例将在3.2.8节讨论。

3.2.1 Coreness

3.2.1 核心性

We classify frame elements in terms of how central they are to a particular frame, distinguishing three levels: core, peripheral, and extra-thematic. A fourth possible value for this attribute, called core-unexpressed is also discussed below. 我们依据框架元素对于框架的重要程度划分等级，分为三级：核心(core)、外围(peripheral)、非题元(extra-thematic)；还有第四种属性core-unexpressed（核心-未实现），下文介绍。

A core frame element is one that instantiates a conceptually necessary component of a frame, while making the frame unique and different from other frames. **核心框架元素**：是框架概念上必不可少的组成部分，正是核心FE让该框架区别于其他框架。

For example, in the Revenge frame, Avenger, Punishment, Offender, Injury, and Injured party are all core frame elements, because an act of revenge necessarily includes these participants. 例如复仇框架中，复仇者、惩罚、冒犯者、伤害事件、受害方全部都是核心框架元素；一次复仇行为概念上必然包含这些参与者。

One cannot imagine an act of revenge that is not preceded by a (perceived) offense or one that is not directed against anybody. 无法想象一次复仇，在它之前不存在（主观认定的）过错，也没有针对任何对象。

In determining which frame elements are considered core, we also consider some formal properties that provide evidence for core status. These properties are typically co-present, although they need not be. 判定一个FE是否属于核心，我们同时参考部分形式特征证据；这些特征一般会同时满足，但不是必须全部满足。

• When an element always has to be overtly specified, it is core. For instance, the verb resemble in the Similarity frame always requires a post-verbal complement NP denoting an entity that is similar to the entity denoted by the subject. • 如果某个成分必须显性出现，则属于核心。例如Similarity（相似）框架的resemble，后面必须接名词短语补语，代表和主语所指对象作对比的实体。

• A frame element which, when omitted, receives a definite interpretation, is also core. For instance, when the verb arrive is used only with a Theme-subject, as in John arrived, a particular Goal location that the Theme reaches still has to be understood in the context. Goal, therefore, must be a core frame element. • 如果某个FE被省略，但解读时必须得到确定指称，它同样是核心FE。例如句子John arrived，只有主题作主语，但语境中必须解读出主题抵达的特定目标地点。因此Goal目标地点是核心FE。

• A frame element whose semantics cannot be predicted from its form, in particular from any marking prepositions, ought to be core. From this, we can derive two corollaries given below. • 如果FE的语义无法从表层形式（尤其标记介词）预判，该FE应当为核心。由此可以得到下面两条推论。

– A frame element that has no formal marking should be core. Thus, frame elements that can be subject or object in a simple active sentence ought to be core since these slots host frame elements of many different kinds and knowing that something is a subject or object is not particularly informative. ——没有任何形式标记的FE属于核心。简单主动句中可以充当主语、宾语的FE都属于核心；因为主语宾语位置可以容纳种类繁多的角色，仅仅知道是主语或宾语无法推知语义角色。

A frame element that has idiosyncratic formal marking should also be core. A good example of this is the prepositional verb depend on. The preposition on does not occur as a marker of the same meaning with predicates in many other frames. ——拥有特异形式标记的FE同样是核心。典型例子是短语动词depend on；介词on不会在大量其他框架中用来标记相同语义。

[4]In its basic spatial sense of ‘in contact with and supported by’, on occurs in many different frames; as a marker of Place or Location frame elements it is totally unremarkable and should not suggest core status for these FEs. [4]on表示基础空间义“接触并被支撑”时，会出现在大量框架；此时标记地点、位置FE，属于普通标记，不能凭此判定FE为核心。

Peripheral FEs mark such notions as Time, Place, Manner, Means, Degree, and the like. They do not uniquely characterize a frame, and can be instantiated in any semantically appropriate frame. **外围框架元素**标记时间、地点、方式、手段、程度这类概念。它们不具备区分框架的独有特征，任何语义适配的框架都可以实例化外围FE。

In respect to the Revenge frame, any report of an event of revenge may also include explicit information about the parameters of time, place, manner, etc. of the revenge, an example of which is given below. 对于复仇框架，描述复仇事件时，可以显性给出复仇发生的时间、地点、方式等参数，见下例。

(8) The bereaved family retaliated [immediately Time]. (8) 遇难者家属[时间 立刻]展开报复。

Extra-thematic frame elements situate an event against a backdrop of another state of affairs, either of an actual event or state of the same type, as illustrated with Iteration, or by evoking a larger frame within which the reported state of affairs is embedded, as shown for Containing event. **非题元框架元素**：把事件放置在另一种事态背景之下。一类如同Iteration迭代，是同类型真实事件/状态作为背景；另一类如Containing event包含事件，唤起一个更大框架，当前描述事态嵌套在该大框架内部。

(9) Thou shalt not exact revenge [twice Iteration] for the same offense. (9) 不可为同一件过错[迭代 两次]实施复仇。

(10) The Aussies took revenge [in a penalty shootout before 2465 fans in Long Beach the next day Containing event]. (10) 澳大利亚队实施复仇，[包含事件 次日长滩，2465名球迷面前的点球大战中]。

Note that extra-thematic frame elements are understood not to conceptually belong to the frames they appear in. We take them to properly be frame elements of other abstract frames that take them as well as the targets that they modify as arguments. 注意，概念层面非题元FE并不归属于它所出现的框架；它们本质属于另外抽象框架的FE，当前目标词只是被该抽象框架拿来充当论元。

Thus, in example (11), we take twice and the verb phrase eat to be arguments of a more abstract Iteration frame. Similarly, in example (12), Lennert, another sweetie in my life, cooked [me] dinner, mmm mmm good. Note that, as shown for (12), the native frame of the extra-thematic frame element need not be evoked by lexical material, it may simply be evoked constructionally. 于是例(11)中twice和动词eat都作为更抽象Iteration框架的论元。同理例(12)中Lennert, another sweetie in my life, cooked [me] dinner，非题元FE所属原始框架不需要由词汇唤起，可以直接由构式唤起。

(11) Learn how to spend a few extra minutes planning complementary menus where you cook once and eat [twice]. (11) 学会多花几分钟规划配套菜单，做一次饭，可以吃[两次]。

(12) Lennert, another sweetie in my life, cooked [me] dinner, mmm mmm good. (12) 伦纳特，我生命里另一个暖心的人，给[我]做了晚饭，味道棒极了。

The view of extra-thematic frame elements presented here entails that these frame elements are necessarily the same across all the ‘host’ frames in which they appear. That is, unlike core and peripheral frame elements, extra-thematic frame elements do not have a frame-specific understanding. 这套对非题元FE的理解意味着：该类FE在所有宿主框架中含义完全统一。和核心、外围FE不一样，非题元FE不存在某个框架专属的解读。

By comparison, although many core frame elements named Agent share properties with each other due to Inheritance and Using relations, they do not necessarily have identical properties. 与之对比，很多名叫Agent施事的核心FE，虽然依靠继承、使用关系共享部分特征，但并不保证全部属性完全相同。

More importantly, one cannot predict the frame role Agent from the fact that an NP has the grammatical function Ext. 更重要一点：仅仅知道一个名词短语拥有Ext外部论元语法功能，无法推导出它的语义角色就是Agent施事。

(For an overview of the most frequently occurring extra-thematic frame elements, the reader is referred to Appendix A.) （高频非题元框架元素总览，参见附录A。）

The value “Core-Unexpressed” is a special notational shorthand. It is assigned to FEs that behave like core frame elements in the frame where they are marked as Core-unexpressed but which, counter to expectation, may not be used for annotation in descendants of that frame. Core-Unexpressed（核心-未实现）是特殊简写

标记。标记该属性的FE，在本框架内具备核心FE的全部特征，但该框架的子继承框架，不能使用这个FE做标注。

We do not consider core-unexpressed frame elements to be violations of our definition of full inheritance. Our reasoning can be illustrated with the example of the core-unexpressed Act frame element in the Intentionally act frame, which is exemplified in (13). Core-unexpressed并不会违背完整继承的定义。我们拿Intentionally act有意行动框架中标记为Core-unexpressed的Act行动FE举例说明。

(13) I'll do [the vacuuming Act]. (13) 我将要做[行动 吸尘打扫]这件事。

In the many child frames of Intentionally act such as Choosing, Perception active, etc., the idea of an Act is as relevant as in the Intentionally act parent frame. However, in the child frames the frame element is absorbed by the lexical units in the frame and cannot be separately expressed. 在Choosing做选择、Perception active主动感知等大量子框架中，Act行动概念同样存在；但子框架内部，该FE被词汇单元吸收，不能单独显性表达。

For example, the sentence *I chose decision the blue one is simply ungrammatical.* 例如句子I chose decision the blue one 本身不合语法。

Marking the frame element Act as core-unexpressed in the Intentionally act parent frame allows us to keep the frames that are lower in the hierarchy from including an inherited FE which for any lexical unit in the frame could at most be annotated on the target itself, but never be expressed separately. 在父框架Intentionally act中将Act标记为core-unexpressed，可以避免下层子框架继承来一个FE：这个FE最多只能标记在目标词本身，永远不能作为独立成分表达出来。

Coreness marking makes the most sense for event and state frames, and in these frames we use all three statuses. In frames whose LUs are artifacts or natural kinds, we only use a two-way distinction (core and peripheral), noting that the values do not have exactly the same meaning as with events. 核心性标记最适合事件框架、状态框架，这类框架会使用全部三种状态。如果框架词汇单元是人造物、自然事物，则只区分core核心与peripheral外围两级；这两级的内涵和事件框架不完全等同。

In such cases, typically there is just one core frame element which is marked on the target word. For instance, in the Clothing frame the FE Clothing is core, and all other FEs are peripheral. 这类框架一般只有一个核心FE，标记在目标词身上。例如Clothing衣物框架中，Clothing是核心FE，其余全部FE都是外围。

3.2.2 Frame element relations

3.2.2 框架元素之间的关系

In providing a semantic analysis of the combinatorial possibilities of our target predicates–rather than one stated only in terms of phrase types and grammatical relations–we have observed that frame elements are not independent of each other. 在描写目标谓词组合可能性的语义分析中（不只是短语类型、语法关系层面描写），我们观察到框架元素之间并不是互相独立。

The frame elements are related to the frame and required by it, as well as interrelated directly in a number of ways. These interrelationships have a direct impact on annotation, as they license the absence of core frame elements, which must normally all be accounted for in every annotation set, or (more rarely) require frame elements that might otherwise be optional. FE既隶属于框架、受框架约束，互相之间还存在多种直接关联。这些相互关系直接影响标注：它们规定哪些核心FE可以缺省（正常情况下每一组标注都要处理全部核心FE）；少数场景下，还强制要求本可以选省的FE必须出现。

The remaining parts of this section discuss the three types of relations between frame elements of the same frame that we recognize, as well as some respects in which the current treatment of frame element sets needs further refinement. 本节剩余部分介绍同一框架内部FE的三类关系，同时说明当前FE集合处理方案尚待完善之处。

3.2.2.1 Coreness Sets

3.2.2.1 核心集合（CoreSet）

In our annotation practice, we often find that some groups of FEs seem to act like sets, in that the presence of any member of the set is sufficient to satisfy a semantic valence of the predicator. We refer to such a group of FEs as a coreness set, or CoreSet. 标注实践中经常观察到部分FE形成集合；集合中任意一个成员显性出现，就足以满足谓词的一条语义配价。我们把这类FE集合叫做CoreSet核心集合。

For instance, Source, Path, and Goal are core FEs in the various motion frames in the database. However, although possible, it is not necessary, and in fact unusual, for all three FEs to co-occur, as in example (14). 例如各类运动框架中，Source起点、Path路径、Goal目标地点都是核心FE。虽然三者可以同时出现，如例句(14)，但不是必须共现，而且三者同时出现其实并不常见。

(14) Fred went [from Berkeley Source] [across North America and the Atlantic Ocean Path] [to Paris Goal]. (14) 弗雷德[起点 伯克利]出发，[路径 横跨北美与大西洋]，抵达[目标地点 巴黎]。

(15) Martha hiked [from Berkeley Source] [to Oakland Goal]. (15) 玛莎徒步，[起点 伯克利]走到[目标地点 奥克兰]。

(16) Elaine walked [to Monterey Goal]. (16) 伊莱恩步行前往[目标地点 蒙特雷]。

(17) I saw Peter sneak [past the guard Path]. (17) 我看见彼得悄悄从守卫身边[路径 经过]。

(18) Juan was walking [out of the office Source] when I arrived. (18) 我抵达的时候，胡安正从办公室[起点 里面]往外走。

FrameNet's normal annotation practice demands that we account for all core FEs and we could keep track of the un-instantiated FEs in the example sentences above by using null-instantiation labels as described in section (3.2.3). FrameNet常规标注要求处理全部核心FE；理论上可以用3.2.3节零实例化标签记录上例中没有实现的FE。

However, we prefer to group the FEs in CoreSets and not mark null instantiation for each member FE in cases where the FEs have an informational and conceptual interdependence. 但是，如果一组FE存在信息层面、概念层面的相互依存，我们优先将它们归为CoreSet，而不对集合内每个成员都打零实例化标签。

Source, Path, and Goal, for instance, are clearly related via a notion that we might call 'full path'. 举例，Source、Path、Goal三者依靠"完整路径"这一概念紧密关联。

By contrast, omission of the Ingestibles for eat in the Ingestion frame is not related to the presence or absence of any other frame element. 与之对比，Ingestion进食框架中Ingestibles摄入物被省略，这件事和其他FE是否出现没有关系。

The sentence I'm eating (with my friends/in the kitchen/now) is acceptable with or without any of the frame elements given in parentheses. 句子I'm eating (with my friends/in the kitchen/now)，括号内成分无论有没有，句子都成立。

3.2.2.2 Requires

3.2.2.2 Requires（要求关系）

In some cases, the occurrence of one core FE requires that another core FE occur as well. To illustrate, in the Attaching frame Item, Goal, and Items all are core FEs. If Item occurs, then Goal is required, as shown below, where the sentence without a Goal is unacceptable. 有些情况下，一个核心FE出现，就强制要求另一个核心FE也必须出现。以Attaching附着框架举例：Item单个对象、Goal目标位置、Items多个对象全部是核心FE。如果Item出现，则Goal必须出现；下面例子可以看到，没有Goal的句子不合法。

(19) The robbers tied [Paul Item] [to his chair Goal]. (19) 劫匪把[单个对象 保罗]绑到[目标位置 他的椅子上]。

(20) * The robbers tied [Paul Item]. (20) *劫匪把[单个对象 保罗]绑了。

In this situation, we mark a Requires relation between the two frame elements. 这种情况下，我们在两个FE之间标记Requires要求关系。

The Requires relation occurs in almost all frames that have a construal alternation between a symmetric construal, when a single frame element name is used, and an asymmetric construal, when two frame elements with names of the are used. Requires关系几乎出现在所有具备对称-非对称构式交替的框架中：对称构式只用一个FE名称；非对称构式使用两个独立FE名称。

Some sample frames are Compatibility, Chatting, and Similarity; there are many more. 例子有Compatibility兼容、Chatting交谈、Similarity相似，还有大量其他框架。

3.2.2.3 Excludes

3.2.2.3 Excludes（互斥关系）

In some cases, if one of the FEs in a group of conceptually related FEs shows up, no other FE from that group can. Again, in the Attaching frame, if Items occurs, then Item and Goal are excluded. 有些场景，一组概念相关的FE中，只要其中一个出现，同组其他FE就不能出现。回到Attaching附着框架：如果Items（多个对象）出现，则Item单个对象、Goal目标地点互斥不能出现。

In this situation, we say that Items excludes Item and Goal. 这时我们标记Items对Item、Goal是Excludes互斥关系。

(21) The robbers tied [his ankles Items] together. (21) 劫匪把[多个对象 他的脚踝]绑在了一起。

The above Excludes-relation in the Attaching frame is an instance of a much more common pattern of alternation between a symmetric/reciprocal and an asymmetric construal of events or states involving two parties. Attaching框架上面这套Excludes互斥，是非常普遍的模式：事件、事态在对称/互反构式与非对称构式之间交替。

In most frames, where the alternation is possible, the names of the frame elements reflect the underlying alternation between reciprocal and asymmetric construal of events or states involving two parties. 大部分具备该交替现象的框架，FE命名本身就会体现这种互反-非对称构式交替。

For instance, in the Similarity frame with lexical units such as similar, different, etc., we have the frame elements Entity 1 and Entity 2, and Entities. Usually, one is allowed to infer equal participation in the event or state by the grammatically less profiled participant (Participant 2). 例如Similarity相似框架，词汇单元有

similar、different等；FE包括Entity1实体1、Entity2实体2、Entities复数实体。通常可以推导出句法侧显程度弱的Participant2，同等参与该事态。

However, in the Attaching frame, the Goal (which would be Item 2 under our normal naming scheme) is not readily understood as itself being secured or immobilized via attachment to the Item on the asymmetric construal, we selected a name that reflects the fact that the usual inference to equal status for Participant 2 is not warranted. 但Attaching附着框架中，非对称构式下的Goal（本该叫Item2）不能被解读为同等参与，即被附着固定；因此我们的命名没有沿用Entity1/Entity2，而是专门命名，体现该推导不成立。

The Excludes relation also manifests in frames where an event can be brought about either by an intentional Agent or by a Cause event. Excludes互斥关系也出现在这类框架：事件既可以由有意向Agent施事引发，也可以由Cause诱因事件引发。

Consider the following examples from the Placing frame. 请看Placing放置框架例句。

(22) [The same flood tide that had brought such a good harvest of tiles Cause] heaped a mass of driftwood onto the Reach. (22) [诱因 带来大量瓦片丰收的同一场涨潮]把一大堆浮木堆积到这片水域。

(23) [Bill Agent] deposited the bag of croissants and the Financial Times carelessly on the hall table. (23) [施事 比尔]漫不经心地把可颂面包袋子与金融时报放在玄关桌上。

The two sentences represent two different construals of Placing scenes. Sentence (22) focuses on an event as causing the change in location of the Theme, whereas sentence (23) focuses on an Agent who through their involvement in an unspecified event, most likely an intentional action involving his hands and body, causes the change of location of the Theme. 两个句子代表放置场景两种不同构式：句(22)聚焦某个事件作为诱因，带来主题位置改变；句(23)聚焦施事，施事通过某个未指明的、大多是身体参与的有意动作，造成主题位置改变。

The two construals are incompatible (since there is only one subject slot) and the frame elements Agent and Cause stand in an Excludes relation to each other. 两套构式互相不兼容（主语位置只有一个）；因此Agent施事与Cause诱因之间是Excludes互斥关系。

[6]In any sentence in which the event that the Agent is involved in is specified together with the Agent, the Agent would have to be realized as a subject and the event as a clause or PP. We then simply label that event a Means rather than a Cause. [6]如果句子同时指明施事以及施事参与的事件，施事占据主语，该事件由小句或PP实现；此时这个事件标记为Means手段，而不是Cause诱因。

Another clear instance of the Excludes relation between frame elements occurs in the Evading frame, where an Evader moves under its own power to thereby avoid Capture or contact with a Pursuer. Evading逃避框架是FE互斥关系又一个清晰案例：逃避者依靠自身行动，躲避抓捕，或是躲避追捕者。

The Capture is an actual or hypothetical event in which the Pursuer takes physical control of the Evader. The Capture frame element and the Pursuer are thus clearly interrelated but only one of them can appear as a dependent of a target in the Evading frame. Capture抓捕是真实或假想事件：追捕者对逃避者取得物理控制。Capture与Pursuer追捕者概念高度关联，但二者只能有一个充当Evading框架目标词的从属成分。

(24) Sheriff's officials said they apprehended a gang member after he evaded [them Pursuer]. (24) 治安官人员称，他们逮捕了一名帮派分子，这名分子之前躲避过[追捕者 他们]。

(25) He had successfully evaded [arrest Capture]. (25) 他成功躲避了[抓捕 逮捕]。

Finally, note that the Excludes relation strictly applies only to the direct syntactic dependents of a target word, that is, to first layer annotation. Frame elements that exclude each other may co-occur in an annotation set if they appear on separate annotation layers. 最后注意：Excludes互斥关系严格只针对目标词直接句法从属成分，也就是第一层标注。如果两个FE分别标记在不同标注层，互斥FE可以同时出现在同一组标注。

(26) Perkins McLain evaded [capture [through Spain Pursuer] Capture] . (26) 帕金斯·麦克莱恩躲避了[抓捕，[追捕者 途经西班牙]]。

(27) The discussions [between [Miller Interlocutor 1] and [the dean Interlocutor 2] Interlocutors] went nowhere. (27) [对话参与者复数：对话参与者1米勒、对话参与者2院长]之间的讨论毫无进展。

3.2.2.4 Future refinements

3.2.2.4 待完善改进点

One problem is that we have no explicit treatment of the idea of proto-frame elements, of which other frame elements are more specialized expressions. 第一个问题：目前没有显式处理**原型框架元素(proto-frame elements)**；其他FE都是原型FE的特化表达。

In cases like (28), we would prefer not to pick either specifically Agent or Cause as constructionally null instantiated, since the context might not provide enough information to resolve that question. Instead we would make reference to a superordinate frame element (call it *Force) that is vague about intentionality and event-person distinction.* 如句*(28)*，我*们*不希望强行判定是*Agent*还是*Cause*做构式零*实*例化，因*为*上下文信息不足以区分二者；理想做法是引用一个上位原型*FE*，命名*为*Force作用力，它不对是否有意向、区分事件还是人物做硬性规定。

(28) The car got damaged while parked outside of our house. (28) 这辆车停在我们家屋外的时候受损了。

(29) A. How's Susan working out? B. She's very good. (29) A：苏珊工作表现怎么样？B：她非常优秀。

Having an explicit representation of proto-frame elements would also be useful in dealing with certain linguistic expressions that seem to instantiate the superordinate proto-frame element rather than one of its more specialized manifestations. 如果原型框架元素有显式表征，对于处理部分语言表达会很有用，这类表达实例化的就是上位原型FE，而不是某个特化子类FE。

Consider the phrase trench to trench in (30): it does not refer to the Source or Goal of a Path but neither does it refer to the middle Path which would be compatible with the specification of a final Goal. 拿句(30)的trench to trench举例：它既不代表路径的Source起点，也不代表Goal终点，同时也不是普通的中间Path路径。

(30) He crawled [trench to trench], looking for some sign of Stephen. (30) 他[战壕到战壕之间]匍匐前进，寻找斯蒂芬的踪迹。

Another use for proto-frame elements involves inheritance relations. In some cases, an inheriting frame will allow only one of the (Excludes or CoreSet group) FE from the parent frame. Superficially, this violates the rule that requires child frames to have a corresponding FE for each core/peripheral FE of the parent. In a deeper sense, it is fulfilling every constraint of the parent, merely adding a constraint that prevents one of the construals possible in the more generic case. 原型FE还可以用于处理继承关系。部分子继承框架，只允许父框架Excludes互斥集合、CoreSet核心集合中的某一个FE。表面上看违背规则：子框架应当为父框架每个核心、外围FE配备对应FE。但从深层语义看，子框架仍然满足父框架全部约束，只是额外增加一条约束，禁止父框架允许的其中一种解读构式。

This would make it possible, for instance, to have our *Murder frame (with only agentive causes) as a child frame of the Killing frame (which allows Causes or Agents).* 举例子：*这样*就可以把Murder谋杀框架（只允许有意向施事）设置为Killing杀害框架（既允许Cause诱因，也允许Agent施事）的子框架。 [7]Currently, predicates denoting intentional killing are in the Killing frame together with predicates that allow either intentional or non-intentional causation. [7]当前版本，表有意杀害的谓词和允许有意、无意致亡的谓词全部放在Killing杀害框架。

A second major problem that remains concerns the treatment of subject selection constructions. The current policy on coreness requires that all frame elements that can occupy the subject position be marked as core frame elements. If fully carried out, this would lead to Means and Instrument frame elements having core status in many frames, and further being part of a CoreSet with Agent, since the two frame elements may co-occur. 第二个遗留大问题是主语选择构式的处理。现行核心性判定规则：凡是可以占据主语位置的FE，标记为核心FE。如果严格执行，很多框架中Means手段、Instrument工具都会被标记为核心FE，并且和Agent施事组成CoreSet核心集合，因为二者可以共现。

However, we have ignored this complication on the assumption that subject selection constructions are explainable through more general constructions. 但我们暂时搁置这个复杂问题；我们的假设是：主语选择构式可以依靠更通用的构式规则来解释。

Consider the verb open in the Closure frame. In (31), we have a canonical Agent subject but in (32) an Instrument fills the subject slot. Note that the frame conceptually requires an Agent; uses of open involving a Cause such as The wind opened the door are handled in another frame. 以Closure闭合框架中的open动词举例：句(31)是标准Agent作主语；句(32)Instrument工具占据主语位置。注意概念层面open仍然预设Agent施事；The wind opened the door（风把门吹开）这类Cause诱因作主语的用法放在另一个框架处理。

(31) [John Agent] opened the door. (31) [施事 约翰]打开了门。

(32) [The key Bill gave him Instrument] opened the door right away. (32) [工具 比尔给他的那把钥匙]立刻打开了门。

Were there a separate way of representing the fact that Instrument subjects are possible, it would not be necessary to give Instrument core status. This would avoid introducing a coreness set of Agent and Instrument. 如果有独立机制刻画"工具可以充当主语"这件事，就不需要把Instrument标记为核心FE，也就不需要设立Agent-Instrument的CoreSet集合。

It would also expose the essential lexical similarity between the English frames and the frames of languages such as Japanese, where subject selection constructions are much more restricted and Instruments and Means rarely, if ever, appear as subjects. 同时也可以更好体现英语框架与日语这类语言框架之间本质词汇相似性；日语主语选择构式限制严格，工具、手段几乎不能充当主语。

3.2.3 Null instantiation

3.2.3 零实例化

Sometimes FEs that are conceptually salient do not show up as lexical or phrasal material in the sentence chosen for annotation. Nevertheless, we indicate their absence since it provides lexicographically relevant information regarding omissibility conditions. 有些概念上凸显的框架元素，在待标注句子中没有以词汇或短语形式表层出现。即便如此，我们仍要标记该角色的缺失，因为这可以提供与可省略条件相关的词典学信息。

With respect to null instantiation, verbal, adjectival, and prepositional targets are treated identically. 针对零实例化现象，动词、形容词、介词类标注目标的处理方式完全一致。

(The noun-target case will be discussed in Section 3.4.3.) （名词目标的情况将在3.4.3节讨论。）

The following examples show omitted elements with each part of speech. 下面的例句展示不同词性目标词对应的被省略框架元素。

(33) That will suffice . [DNI Standard] (Sufficiency) (33) 这就足够了。[DNI 标准]（充足框架）

(34) The result should be similar . [DNI Entity 2] (Similarity) (34) 结果应当是相似的。[DNI 实体2]（相似框架）

(35) I tried to put the toys back in . [DNI Ground] (Locative relation) (35) 我试着把玩具放回去。[DNI 背景参照物]（方位关系框架）

Not all cases of frame element omission are alike. We recognize three different cases, one that is not lexically specific and two that are. 框架元素省略分为不同类型。我们区分三类：一类不属于特定词汇专属现象，另外两类为词汇专属现象。

We will now discuss the three types of omission, focusing primarily on verbs, where the phenomenon is clearest. 下面讨论这三类省略，重点以动词为例，动词场景下该现象表现最清晰。

3.2.3.1 Definite Null Instantiation (DNI)

3.2.3.1 确定式零实例化（DNI）

Cases of definite null instantiation are those in which the missing element must be something that is already understood in the linguistic or discourse context. 确定式零实例化：被省略的成分必须在语言上下文或篇章语境中已经明确可及。

In the following example, the Offender is not expressed overtly in the syntax, but its referent has to be known in the context. 下例中，Offender（冒犯者）没有在句法表层实现，但其指称对象必须依靠上下文获知。

(36) [The monkey Avenger] avenged [himself Injured party]. [Offender DNI] (36) [复仇者 那只猴子]为[受害方 他自己]报了仇。[冒犯者 DNI]

3.2.3.2 Indefinite Null Instantiation (INI)

3.2.3.2 不定式零实例化（INI）

The indefinite cases (sometimes also referred to as existential) are illustrated by the missing objects of verbs like eat, sew, bake, drink, etc., that is, cases in which these ordinarily transitive verbs can be spoken of as used intransitively. 不定式零实例化（也叫存在式）典型案例是 eat、sew、bake、drink 等动词省略宾语；也就是及物动词作不及物用法。

(e.g. Molly rarely eats alone; Granny begins baking about a month before Christmas; Michael even drinks heavily on weeknights.) （例如：莫莉很少独自吃饭；奶奶会在圣诞节前大约一个月就开始烘焙；迈克尔甚至工作日晚上也酗酒。）

As is well known, there are often special interpretations of the missing objects. 众所周知，被省略的宾语往往带有特定解读。

With eat the missing entity is likely to be a meal, with bake it is likely to be flour-based foods, with drink it is likely to be alcoholic beverages, etc. eat省略的对象一般是一餐饭；bake省略对象一般是面食；drink省略对象一般是酒精饮品，诸如此类。

In contrast to anaphoric omissions, with existential omissions, the nature (or at least the semantic type) of the missing element can be understood given conventions of interpretation, but there is no need to retrieve or construct a specific discourse referent. 和回指类省略不同，不定式省略依靠解读规约就能知道被省略成分的性质（至少是语义类型），但不需要从上下文找回或构建某个特定篇章指称对象。

For example, in the Revenge frame, all lexical units happen to allow the frame element Punishment to be omitted under indefinite null instantiation. 举例：复仇框架下，所有词汇单元都允许Punishment（惩罚）角色以INI不定式零实例化省略。

This is shown for avenge in (37). 例句(37)以avenge展示该现象。

(37) He took it out on Scarlet in the same way as [he Avenger] avenged [himself Injured party] [on her Offender] [for the pressures at work and the demands of his first wife Injury] . [INI Punishment] (37) 他就像[复仇者 他]为[受害方 自己]、因[伤害事件 工作压力与第一任妻子的种种要求]向[冒犯者 她]复仇那样，把怒火发泄在斯嘉丽身上。[INI 惩罚]

Note that both in the case of definite and indefinite null instantiation, the LUs in a frame may differ from each other in whether or not they allow the omission. 注意：无论是DNI还是INI，同一框架内不同词汇单元，对该角色是否允许省略的规定可以不一样。

For instance, while eat allows its object to be omitted, devour does not, even though they are both in the Ingestion frame. 例如同属Ingestion进食框架，eat允许宾语省略，但devour（狼吞虎咽）不允许。

Verbs that usually require an argument to be present (or only allow it to be omitted under conditions of definiteness) can be used in a generic construction with indefinite null instantiation, as shown below. 通常要求论元显性出现，或者只允许DNI省略的动词，在泛指构式中也可以使用INI不定式零实例化，见下例。

(38) He takes and never gives back. [INI Theme] (38) 他一味索取，从不归还。[INI 主题]

3.2.3.3 Constructional Null Instantiation (CNI)

3.2.3.3 构式式零实例化（CNI）

Constructionally omitted constituents (also called structurally omitted) have their omission licensed by a grammatical construction in which the target word appears, and are therefore more or less independent of the LU. 构式式零实例化：成分省略由目标词所处的语法构式授权，基本和词汇单元本身属性无关。

Cases of CNI include: the omitted subject of imperative sentences, the omitted agent of passive sentences, the omitted subjects of independent gerunds and infinitives (i.e., the PRO-elements of generative grammar), and so on. CNI的例子包括：祈使句省略主语、被动句省略施事、独立动名词/不定式省略主语（生成语法中的PRO）等等。

In each of the following two examples, the FE Avenger is tagged with the symbol CNI. 下面两个例句，Avenger复仇者都标记CNI。

(39) Family feuds last for generations, and [slurs on honor Injury] are avenged [by murder Punishment] . [CNI Avenger] (39) 家族世仇世代延续，[伤害事件 对名誉的诋毁]会以[惩罚 杀戮]的方式得到报复。[CNI 复仇者]

(40) Get even [with her Offender] [for this Injury] . [CNI Avenger] (40) 向[冒犯者 她]为[伤害事件 这件事]讨回公道。[CNI 复仇者]

[8]Note that the name of the Injured party (the FE is shown as Victim in the figure, but has been renamed to Injured party), Pedro, forms part of the FE Injury; see Sec.3.2.5 below. [8]注意：图中该框架元素显示为Victim受害者，现已改名为Injured party受害方；Pedro属于Injury伤害事件框架元素的一部分，参见下文3.2.5节。

In addition, we use CNI for missing objects in instructional imperatives such as exemplified below, even though the omission is not dependent on a particular construction, but rather on a particular genre, such as cookbooks and product labels. 另外，菜谱、产品说明书这类语体下的指令祈使句，宾语被省略，同样标记CNI，即便这种省略不完全由某一条语法构式决定。

(41) Cook on low heat until done. [CNI Food] (41) 小火烹煮直至熟透。[CNI 食物]

(42) Tie together loosely. [CNI Items] (42) 宽松地捆扎到一起。[CNI 多个对象]

The experiential perfect also licenses object omissions that are not possible in simple assertions of the experiential perfect. 经历完成体也允许宾语省略，这类省略在简单陈述句中一般不成立。

(43) Have you ever fostered [CNI Child] before? (43) 你以前有没有收养过孩子？[CNI 孩子]

Note that particular constructions licensing argument omission specify particular interpretations, either indefinite (existential) or definite (anaphoric). 注意：授权论元省略的特定构式，会指定解读类型，要么不定（存在式），要么确定（回指式）。

For instance, the instructional imperative construction in (41) and (42) specifies a definite interpretation, whereas the experiential perfect exemplified in (43) specifies an indefinite interpretation. 例如(41)(42)指令祈使句给出确定解读；(43)经历完成体给出不定解读。

The CNI label thus collapses the interpretational distinction that we make among the lexically licensed omissions; it does not represent a separate kind of interpretation in addition to the definite and indefinite types. CNI标签把词汇层面区分的两种解读合并处理；它本身并不是DNI、INI之外的第三种独立解读类型。

3.2.4 Incorporation

3.2.4 并入

There are many frames containing verbs which generally involve a particular of frame element, but where some of the verbs incorporate information about that frame element in their definition. 很多框架中的动词，概念上会涉及某个特定框架元素，但其中部分动词会在词义内部把该框架元素的信息"并入"词义。

Consider, for example, verbs of body movement; typically certain verbs are expected to co-occur with the name of a body part, even when the identity of the body part is clear from the meaning of the verb. 以身体动作类动词举例：这类动词通常期待搭配身体部位名词，但有些动词词义本身就已经明确身体部位。

A dog wags its tail, people arch their brows, bat their eyes, purse their lips, etc. 狗摇尾巴，人扬起眉毛、眨眼睛、撅嘴唇等等。

But in the case of smile, grimace, frown, pout, and scowl, the affected body part is not separately expressed; we say that it is incorporated. 但smile微笑、grimace做鬼脸、frown皱眉、pout撅嘴、scowl怒视，受影响的身体部位不会单独说出来；我们称该角色被并入词义。

Some verbs in this frame can optionally express the expected body part: one can say either She blinked or She blinked her eyes. 该框架部分动词可以选择是否把并入的身体部位显性表达出来；既可以说 She blinked，也可以说 She blinked her eyes。

Likewise, in the Placing frame, many verbs incorporate the Goal FE (i.e. the place where the Theme ends up) such as bag.v, bin.v, bottle.v, box.v, cage.v, crate.v, file.v, garage.v. 同样，Placing放置框架中大量动词把Goal目标地点FE并入词义：bag装进袋子、bin放进收纳箱、bottle装瓶、box装箱、cage关进笼子、crate装入板条箱、file归档、garage送入车库。

Note that it is still possible to further specify the incorporated FE explicitly, as in They bottled the wine in custom-made blue bottles, in which case, in custom-made blue bottles is annotated as the Goal FE, as usual. 注意并入的FE依然可以额外显性指明。例如句子 They bottled the wine in custom-made blue bottles，此时 in custom-made blue bottles照常标记为Goal目标地点框架元素。

3.2.5 Frame element conflation

3.2.5 框架元素融合

In some cases, information about two frame elements is expressed in a single constituent, a situation we call conflation. 有些情况下，同一个句法成分承载两个框架元素的信息，该现象叫做框架元素融合。

For instance, the concept of ousting somebody from office requires an understanding of the incumbent of the office and the identity of the office; each can be represented separately in a sentence like We ousted Jones as mayor. 举例：罢免某人职务，概念上同时涉及在职人员和职位本身；可以分开表达：We ousted Jones as mayor（我们罢免了琼斯的市长职务）。

But in We ousted the mayor, the direct object stands for both the office and the incumbent. 但句子 We ousted the mayor，直接宾语同时既代表职位，也代表担任该职位的在职者。

We also find examples of frame element conflation in the Revenge frame. In particular, the Injured party may be contained as a possessive in the phrase that realizes the Injury frame element, as seen in the following example: 复仇框架同样存在框架元素融合。特别是Injured party受害方，可以作为领属成分包含在实现Injury伤害事件的短语之中，如下例：

(44) [He Avenger] avenged [Pedro’s death Injury] [by taking out the poker-faced Guards Officer Punishment].
(44) [复仇者 他]为[伤害事件 佩德罗之死]复仇，[惩罚 除掉那位面无表情的警卫军官]。

Here, the possessive Pedro’s realizes the frame element Injured party, the person who suffered the Injury. 本句中领属词 Pedro’s 实现了Injured party受害方框架元素，也就是承受伤害的人。

In such cases, the annotation tool allows for the creation of an additional FE layer, enabling the secondary annotation of (parts of) constituents in the same frame. 遇到这种情况，标注工具可以新建一层FE标注层，对同一成分（或成分片段）做第二层标注。

Note that there is never a phrase type or grammatical function indicated for the frame elements on the secondary FE layer. 注意：第二层FE层上的框架元素，永远不会搭配短语类型、语法功能标签。

Even with conflation, it is still possible for the FE annotated on the secondary layer to be instantiated as a constituent of its own in the syntax. 即便发生框架元素融合，第二层标注的FE仍然可以在句法中作为独立成分显性出现。

In example (45), the information about the person who was hit is encoded by the direct object me, as well as by the possessive determiner my inside the preposition on. 例(45)中，被击中的人，既由直接宾语 me 编码，也由介词on内部的领属限定词 my 编码。

(45) He hit me on my hand. (45) 他打在了我的手上。

In such cases, we tag the possessive on the second layer with the same frame element label that is applied on the first annotation layer to the object of the verb. 这种情况下，我们把领属词在第二层打上和动词宾语第一层完全相同的FE标签。

3.2.6 Syntactic locality

3.2.6 句法局部性

In general, we select sentences for annotation where, with the exception of subjects, all frame elements are realized by constituents that are part of the maximal phrase headed by the target word. 总体来说，我们挑选标注例句遵循：除主语之外，全部框架元素，都由目标词所统领最大短语内部的局域成分实现。

In addition, there are several clear cases, typically involving subjects, where the combination of particular lexical items and grammatical structures guarantees a particular interpretation of phrases which are not dominated by the maximal phrase headed by the target. 但有几类明确例外，大多和主语相关：特定词汇搭配特定句法结构，可以让不受目标最大短语统制的短语获得确定解读。

The first case in which we annotate non-local constituents consists of cases in which the target word is syntactically governed by a raising or control predicate. 第一类需要标注非局域成分的场景：目标词在句法上被提升动词或者控制谓词支配。

In such cases, the valence properties of the higher raising or control predicate guarantee that one of its arguments is also interpreted as an argument of the target, even though the relevant argument is not dominated by the maximal phrase headed by the target. 此时高层提升/控制谓词的配价属性，保证它的某个论元同时被解读为目标词的论元，即便该论元不在目标词的最大短语之内。

The sub-examples below illustrate raising and control predicates. 下面子例句展示提升动词与控制谓词。

Raising

提升动词

Subject to object 主语到宾语提升

(46) We expect [John Avenger] to retaliate [against us Offender]. [INI Punishment] [DNI Injury]. (46) 我们期待[复仇者 约翰]展开报复[针对我们 冒犯者]。[INI 惩罚][DNI 伤害事件]。

Subject to subject 主语到主语提升

(47) [John Avenger] seems to have avenged [the death of his brother Injury] [by luring Smithers into a trap Punishment]. (47) [复仇者 约翰]似乎已经为[伤害事件 兄弟之死]完成复仇，[惩罚 将史密瑟斯诱入圈套]。

Control

控制谓词

Subject control 主语控制

(48) [They Avenger] are hoping to get even [with Smithers Offender] [for the insult Injury]. (48) [复仇者 他们]希望向[冒犯者 史密瑟斯]为[伤害事件 这次侮辱]讨回公道。

Object control 宾语控制

(49) The commander ordered [the troops Avenger] not to retaliate [against the rebels Offender]. (49) 指挥官命令[复仇者 部队]不要向[冒犯者 反叛者]实施报复。

Tough-movement

Tough移位

(50) [The defeat Injury] was difficult to avenge. [CNI Avenger] (50) [伤害事件 这次战败]，很难为之复仇。[CNI 复仇者]

Note that Raising and Control cases are not restricted to verbal controllers, i.e. nouns may also serve that function, as illustrated here. 注意提升、控制现象不限于动词控制器；名词同样可以充当控制器，见下面例子。

(51) Only a short few weeks ago, even [my Perceiver] hope of seeing [her Phenomenon] was just a dream. [Subject control] (51) 就在短短几周前，就连[经验者 我]想要见到[现象 她]的希望，还只是一场梦。【主语控制】

(52) Meanwhile, today, Americans, hungering for victory, are puzzling over the Pentagon's order to [the troops Agent] not to put [the Stars and Stripes Theme] [on their vehicles Goal]. (Object Control) (52) 与此同时，渴望胜利的美国人，正对五角大楼下达给[施事 部队]的命令感到困惑：不要把[主题 星条旗]放置在[目标地点 他们的车辆上]。【宾语控制】

(53) The testers gave it a clear thumbs up both for [its Created entity] ease of assembly and sail performance. [Tough-movement] (53) 测试人员给出明确好评，无论是[创造物 它的]组装简易度，还是航行性能。【Tough移位】

The second case in which we annotate clearly non-local constituents with frame element labels concerns targets that occur inside relative clauses. 第二类标注明显非局域成分的场景：标注目标出现在关系小句内部。

Here our policy is to tag not only the constituent containing the relativizer (if there is one) as a frame element but also repeat the FE/GF/PT triple on the antecedent, and further mark the relative word and the antecedent phrase on the Other layer. 我们的处理规则：不仅要把包含关系代词的成分标记FE-GF-PT三元组；还要在先行词上重复这套三元组，并且在Other层标记关系词与先行短语。

Thus, our annotation for the simplest cases with an overt relativizer is as shown in Figures 3.4-3.5. 因此，带有显性关系代词的最简案例标注结果如图3.4-3.5所示。

> Figure 3.4: Annotation of a target in a that-relative clause 图3.4：that引导的关系小句内部目标词标注

> Figure 3.5: Annotation of a target in a which-relative clause 图3.5：which引导的关系小句内部目标词标注

Note that, in contrast to the examples above, the relative word is not always by itself a phrasal constituent. 注意：和上面例子不同，关系词本身不一定独立成为短语成分。

Similarly, the antecedent phrase may be rather complex, as shown in Figures 3.6-3.7. 同理先行短语本身也可以相当复杂，见图3.6-3.7。

When a target occurs in a relative clause without an overt relativizer, as in Figure 3.8, we only annotate the antecedent phrase and mark it with the label Ant on the Other layer. Since we do not assume any kind of zero or non-overt relativizer, the label Rel is not applied anywhere on the Other layer. 如果目标出现在无显性关系代词的关系小句中（图3.8），只标注先行短语，在Other层打上Ant先行词标签。我们不假设存在零形式关系代词，因此不会打Rel关系词标签。

If a relative phrase is governed by a preposition, we end up with identical FE/PT/GF triples applied to the antecedent phrase and the prepositional phrase containing the relative word; the Ant and Rel labels are applied as usual. 如果关系短语受介词支配，则先行短语和包含关系词的介词短语使用完全相同FE-PT-GF三元组；照常打上Ant、Rel标签。

In cases with preposition stranding, the same FE-GF-PT triple is assigned to the antecedent, the wh-word (if present), and the stranded preposition. 介词悬置的场景，先行词、疑问词（如有）、悬置介词，分配同一套FE-GF-PT三元组。

> Figure 3.9-3.10: relative-clause preposition-stranding annotation 图3.9-3.10：关系小句介词悬置标注

Finally, infinitival relative clauses with an overt relativizer are treated just like finite relative clauses, which is illustrated in Figure 3.12. 最后，带有显性关系词的不定式关系小句，处理方式等同于限定式关系小句，见图3.12。

Infinitival relative clauses without relativizer, by contrast, receive no Ant-Rel marking at all. For instance, books to read over the break the head that the relative clause modifies is not given the Ant label. 与之相反，不带关系词的不定式关系小句完全不打Ant-Rel标签。例如 books to read over the break，被修饰的中心词books不会标记Ant。

3.2.7 Governing verbs of target nouns, adjectives, and prepositions

3.2.7 目标名词、形容词、介词的支配动词

When annotating verbal targets, we do not record any predicates that may govern them. 标注动词目标时，我们不会记录支配它的上层谓词。

However, in the case of target nouns, adjectives, and prepositions, we want to keep track of several special classes of syntactic governors for lexicographic reasons. 但对于名词、形容词、介词目标，出于词典编纂需要，我们需要记录几类特殊句法支配成分。

The special governors are: Support predicates, Copulas, Controllers, and X-Governors. 特殊支配成分分为四类：支撑谓词(Supp)、系动词(Copula)、控制器(Ctrlr)、X-支配成分(Gov-X)。

The motivation for doing this is as follows. 这么做的动机如下。

It would be theoretically justifiable to omit selecting phrases outside of the standard subcategorization frame of the target noun and to rely on automatic tools for syntactic analysis to identify phrases outside of the target's maximal projection that give information about the filler of a frame element role. 理论上完全可以忽略名词标准次范畴框架之外的短语，依靠句法分析工具去找出目标最大投射之外、提供框架元素填充信息的短语。

However, since one of our goals is to document the phrases capable of satisfying particular FE requirements of the analyzed words, our decision was to increase the scope of our annotation instead. 但项目目标之一就是记录可以满足目标词框架元素要求的各类短语，因此我们选择扩大标注覆盖范围。

A welcome by-product of this decision is that the FN database can also serve as a resource for identifying the Support verbs and prepositions, Copulas, Controllers, and X-Governors that FN annotators often find accompanying particular noun, adjective, and preposition targets. 该做法带来一个有益副产品：FrameNet数据库可以作为资源，用来识别名词、形容词、介词目标经常搭配的支撑动词/介词、系动词、控制器、X-支配成分。

3.2.7.1 Support predicates

3.2.7.1 支撑谓词（Supp）

As noted at the beginning of section 3.2, when the syntactic and semantic head of a clause diverge and the noun (or adjective) target is the semantic head of the phrase larger than its maximal projection, the governing verb or preposition is called a Support, abbreviated Supp. 如3.2节开头所述，小句句法中心和语义中心不一致，名词/形容词目标是更大短语的语义中心，支配它的动词或介词就叫做支撑谓词，缩写Supp。

Four conditions for a Supp: 支撑谓词满足四个条件：

1. the Supp syntactically governs the target,
2. Supp在句法上支配标注目标；
3. the target denotes a state, event, or relation by itself or in combination with other predicates,
4. 目标本身就指称状态、事件或者关系；
5. The Supp has little independent meaning on its own, i.e. the combination of support and noun derives most of its meaning from the noun;
6. Supp自身独立语义很弱；支撑谓词+名词的组合，语义主要来自名词；
7. The Supp is idiosyncratically selected by the frame-evoking target element.
8. 支撑谓词是被唤起框架的目标成分特异选择。

Note: Not all Supp are completely semantically bleached; some add aspect, causation or perspective shift. 注意：不是所有支撑谓词语义都完全虚化；部分会增加体、致使、视角转换语义。

Examples for support-verb constructions with revenge.n from the Revenge frame: 复仇框架中 revenge.n名词的支撑动词例句：

(101) The Americans must have felt as if he was [taking Supp] revenge on them for what had happened. (101) 美国人一定感觉，仿佛他正在【Supp 实施】复仇，针对他们此前的所作所为。

(102) King Menephta [took Supp] awful revenge on a Libyan army he defeated around 1300 BC. (102) 梅内普塔国王【Supp 实施】了残酷的报复，针对公元前1300年左右被他击败的利比亚军队。

In these examples, the noun revenge is the semantic head, while take is marked as Supp on the Noun-layer. 上例中名词revenge是语义中心；take在Noun词性层标记Supp。

The subject of take is annotated as Avenger relative to the target revenge, not relative to take. take的主语，相对于目标词revenge标记为Avenger复仇者，而不是相对于take动词。

Support verb + noun constructions are not equivalent to idioms. 支撑动词+名词不等于习语。

True idioms have non-compositional meaning where you cannot derive the whole meaning piece-by-piece from parts. 真正的习语语义非组合性，整体含义不能逐词推导。

Support-verb constructions are mostly compositional; the noun contributes the core event semantics. 支撑动词结构大多满足组合性，名词贡献事件核心语义。

However, some support verbs are idiosyncratically selected by nouns, which makes them lexicographically relevant. 但支撑动词是名词特异选择，因此具备词典学研究价值。

For example: pose an argument vs make an argument. 举例：pose an argument、make an argument，同样是argument，但搭配不同支撑动词。

Support prepositions are prepositions that combine with noun targets to yield predicative PPs, e.g. in possession, on loan. 支撑介词：和名词目标结合，生成可以作谓词的介词短语，例如 in possession（拥有）、on loan（出借）。

(106) The painting [is Cop] [on Supp] loan from Mr Smithers. (106) 这幅画【Cop 系动词】【Supp 处于】史密斯先生出借状态。

3.2.7.2 Copulas

3.2.7.2 系动词(Cop)

Be, seem, appear etc. are copular verbs. They are a subtype of support predicates with highly bleached semantics. be、seem、appear等是系动词，属于语义高度虚化的一类特殊支撑谓词。

They appear in constructions of form NP_1 Verb NP_2 / AJP / PP. 它们出现在句型 NP_1 + 系动词 + NP_2 /形容词短语 /介词短语。

(114) Jeanne Hossenlopp will be the next chairperson of the department. (114) 让娜·霍森洛普将担任该系下一任系主任。

Copula label Cop is marked on Noun / Adj / Prep layer for non-verbal targets. Cop标签打在名词、形容词、介词专属层，仅用于非动词目标。

3.2.7.3 Controllers (Ctrlr)

3.2.7.3 控制器（Ctrlr）

Controller predicates introduce a separate event from the event denoted by the target noun/adjective, but share one participant argument with the target's frame. 控制器谓词引入独立于目标名词/形容词的另一个事件，但有一个参与者论元会和目标框架共享。

Examples: deserve, offer, encourage, consider, find. 例子：deserve值得、offer提供、encourage鼓励、consider认为、find觉得。

(124) In this case it means that the bad deed deserves revenge. (124) 在这种情况下，就意味着该恶行应当得到报复。

Here deserve introduces a judging event; revenge denotes the retaliation event. The bad deed is shared participant. 本句deserve引入评判事件；revenge代表报复事件；the bad deed是二者共享参与者。

The shared participant constituent is annotated as a frame element relative to the noun target; the verb deserve gets label Ctrlr on Noun layer. 共享参与者成分，相对于名词目标标注框架元素；动词deserve在Noun层标记Ctrlr。

3.2.7.4 X-Governors (Gov-X)

3.2.7.4 X-支配成分（Gov-X）

X-Governor annotation is for artifact / natural-kind nouns that only act as slot-fillers (FE fillers) rather than evoking their own frame. Gov-X标注针对人造物、自然类名词；这类名词只充当槽填充项（框架元素填充者），本身不唤起专属框架。

The verb is marked Gov; the NP headed by artifact noun is marked X. 支配动词标记Gov；人造物名词作中心词的名词短语标记X。

(156) He [built Gov] [two tall towers X]. (156) 他【Gov 建造】【X 两座高塔】。

This annotation records which predicates take the artifact noun as their frame-element filler. 该标注用来记录哪些谓词把该人造物名词用作框架元素填充项。

3.2.8 Discontinuous frame elements

3.2.8 非连续框架元素

A discontinuous frame element means one single FE realization split into two or more separate surface constituents. 非连续框架元素：同一个框架元素的实现形式，在表层拆分成两个或多个独立句法成分。

Multiple separate instances of the same FE (e.g. multiple Path segments) are not discontinuous FEs. 同一个框架元素的多次独立实例化（例如多段路径），不属于非连续框架元素。

Sources of discontinuity in FrameNet annotation: FrameNet标注中非连续现象来源：

1. Raising predicates;
2. 提升谓词；
3. Prepositional passive construction;
4. 介词被动构式；
5. Wh-extraction / topicalization / relativization (preposition stranding);
6. wh移位 /话题化 /关系从句（介词悬置）；
7. Heavy-NP shift / appositive splitting;
8. 重名词短语移位 /同位语拆分；
9. Depictive split modifiers for adjectives / degree constructions.
10. 描绘性拆分修饰语，形容词、程度构式。

(74) [The reader NP.Ext] might [want to excoriate the paper for its coverage of the so-called Clear Skies policy VPbrst.Dep]. (74) [外部论元 读者 NP.Ext] 可能【Dep 裸动词短语】想要痛斥该报纸对所谓"晴空政策"的报道。

(76) [The Peacock Throne NP.Ext] has been sat [on PP.Dep] by the Iran monarchy since the Nadir Shah. (76) [外部论元 孔雀王座 NP.Ext]，自纳迪尔沙时代起，就被伊朗君主坐在【Dep 介词短语 on】上面。

In discontinuous cases, each fragment receives the identical FE label, plus appropriate GF and PT. 非连续场景下，每一个片段都分配完全相同的FE标签，搭配对应的语法功能、短语类型。

3.3 Annotation with verbs as targets

3.3 以动词作为标注目标

Verbs are the most common target type in FrameNet. 动词是FrameNet中最常见的标注目标类型。

3.3.1 Easy cases

3.3.1 简单案例

Easy case: All core frame elements are realized as immediate syntactic dependents of the target verb. 简单案例：全部核心框架元素，都作为目标动词的直接句法从属成分实现。

Each dependent constituent receives triple: FE label, GF label, PT label. 每一个从属成分获得三元组：FE框架元素、GF语法功能、PT短语类型。

Software can auto-assign GF and PT; annotators manually correct mistakes. 软件可以自动分配GF、PT标签；标注人员人工修正错误。

Non-core FEs (peripheral, extra-thematic) also get annotated when present in the sentence. 句子中出现的非核心FE（外围、非题元）同样做标注。

(93) They wake you up [to give you a sleeping pill Purpose]. (93) 他们把你叫醒，【目的 为了给你吃安眠药】。

3.3.2 Expletives

3.3.2 虚设成分

Expletive it / there are non-referential dummy elements; they never instantiate any FE. 虚设成分 it、there 属于无指称占位成分，永远不会实例化任何框架元素。

We mark them with label Null on the Other-layer, no FE / GF / PT. 我们只在Other层标记Null；不分配FE、GF、PT标签。

Typical expletive constructions: subject extraposition, object extraposition, existential there-construction, weather-verb subjects. 典型虚设构式：主语外置、宾语外置、there存现句、天气动词主语。

(94) [It Null] is clear that we won’t finish on time. (94) 【Null 虚设it】很清楚，我们无法按时完成。

(96) [There Null] are more hats in the closet. (96) 【Null 虚设there】衣柜里面还有更多帽子。

3.3.3 Aspect

3.3.3 体

Particles that purely signal aspect are tagged Aspect on the Verb-layer. 单纯表达体意义的小品词，在Verb词性层打上Aspect标签。

(98) They were chattering [away Aspect] in the kitchen. (98) 他们在厨房喋喋不休【Aspect 体小品词 away】。

If particle + verb forms a lexical unit (phrasal verb idiom), particle is part of target, no separate Aspect tag. 如果小品词+动词本身构成一个词汇单元（习语式短语动词），小品词属于目标词一部分，不再单独打Aspect标签。

(100) It's amazing what they can do these days and luckily the sick pulled through. (100) 如今他们能做到的事情令人惊叹；幸运的是病人挺过来了。

> pull through is one single LU, particle through belongs to target. pull through 是完整词汇单元，through属于目标词一部分。

3.4 Annotation with nouns as targets

3.4 以名词作为标注目标

Noun targets include event nouns, relational nouns, artifact / natural-kind nouns. 名词目标分为事件名词、关系名词、人造物/自然类名词。

Event nouns and relational nouns frequently evoke frames; artifact nouns mostly act as slot fillers. 事件名词、关系名词经常唤起框架；人造物名词大多充当槽填充项。

3.4.1 Special governors

3.4.1 特殊支配成分

Four governor classes for noun targets: Support predicates, Copulas, Controllers, X-Governors. 名词目标四类支配成分：支撑谓词、系动词、控制器、X-支配成分。

3.4.1.1 Support expressions

3.4.1.1 支撑表达式

Support verb + event-noun: verb is syntactic head, noun is semantic head. 支撑动词+事件名词：动词是句法中心，名词是语义中心。

Support verbs are selected idiosyncratically by noun LU. 支撑动词由名词词汇单元特异选择。

Categories of support meaning: plain vanilla, aspectual, point-of-view, registral, causative. 支撑动词语义类别：基础型、体变化型、视角转换型、语体差异型、增加致使语义型。

(104) John and I [had Supp] a terrible argument last night. (104) 我和约翰昨晚【Supp 进行】激烈争吵。

(105) John [made Supp] a convincing argument that the project should be funded. (105) 约翰【Supp 提出】有说服力的论证，该项目应当获得资助。

Support prepositions combine with nouns to make predicative PPs: in possession, on loan, at risk. 支撑介词搭配名词生成可作谓语的介词短语。

3.4.1.2 Copulas

3.4.1.2 系动词

Copulas (be / seem / appear etc.) are marked Cop on Noun / Adj / Prep layer. 系动词be/seem/appear等，在名词/形容词/介词专属层标记Cop。

FrameNet no longer distinguishes predicational vs equational copular sentences in annotation labels. FrameNet标注标签层面，不再区分述谓型系词句与等同型系词句。

3.4.1.3 Existential sentences

3.4.1.3 存现句

There-existential constructions: verb (be / ensue / occur) gets Cop / Supp label; expletive there marked Null on Other-layer. there存现构式：be、ensue、occur这类动词标记Cop或者Supp；虚设there在Other层标记Null。

(120) There [was Cop] an argument between the players. (120) 球员之间【Cop 存在】一场争吵。

3.4.1.4 Controllers

3.4.1.4 控制器

Controller verbs introduce separate event, but share one participant with noun-target's frame; verb tagged Ctrlr on Noun-layer. 控制器动词引入独立事件，但和名词目标框架共享一个参与者；动词在Noun层标记Ctrlr。

(126) The Prime Minister encouraged revenge against Absurdistan. (126) 首相鼓励对阿布斯迪斯坦采取报复行动。

encouraged is Ctrlr; revenge is noun target. encouraged为控制器Ctrlr；revenge是名词标注目标。

3.4.2 Noun Compounds

3.4.2 名词复合词

Two treatments for noun-noun compounds: 名词-名词复合词两套处理方案：

1. Compound treated as one single lexical unit (e.g. firing squad.n). No internal decomposition.
2. 复合词整体作为一个完整词汇单元，不做内部拆解，例firing squad行刑队。
3. Decomposed compound: head noun is target; modifier noun is annotated as FE relative to head noun.
4. 可拆解复合词：中心名词作为标注目标；修饰名词相对于中心名词标注框架元素。

Example: fertility rite.n → target rite; fertility annotated Descriptor FE. 例子：fertility rite生育仪式，目标词rite；fertility标记为Descriptor描述词FE。

Modifiers are tagged as Phrase-Type N (non-maximal nominal). 修饰成分短语类型标记N（非最大名词性成分）。

> Important rule: We never annotate head relative to modifier inside compound nouns. 重要规则：复合词内部，永远不以修饰语为目标去标注中心名词。

3.4.3 Null Instantiation with Noun Targets

3.4.3 名词目标的零实例化

Noun-target null-instantiation behaves differently from verb targets. 名词目标的零实例化行为和动词目标不一样。

Quantification and generic reading block DNI for noun targets. 量化、泛指解读，会阻止名词目标使用DNI确定式零实例化。

Only two scenarios we apply DNI for noun targets: 名词目标下，只有两种场景使用DNI标记：

1. Specifically-referring relational nouns (kinship nouns etc.).
2. 指称特定对象的关系名词（亲属名词等）。
3. Event nouns inside finite-clause support-verb constructions referring to a specific single event.
4. 出现在限定小句支撑动词构式中，指称某一次特定事件的事件名词。

Generic / quantified nominal uses do NOT get DNI tags. 泛指、量化的名词用法，不打DNI标签。

(129) Revenge is sweet. (129) 复仇是甜蜜的。

> Generic reading, no DNI applied for Avenger / Offender / Injury. 泛指解读，Avenger、Offender、Injury不使用DNI标记。

(132) The [brother Alter] was found not guilty [DNI EGO]. (132) [关系另一方 这位兄弟]被判无罪。[DNI 本我]

> kinship relational noun case for DNI 亲属关系名词，使用DNI案例。

3.5 Annotation with adjectives as targets

3.5 以形容词作为标注目标

Adjective targets occur both in attributive and predicative position. 形容词目标可以出现在定语位置，也可以出现在谓语位置。

Copula verbs with adjective targets are marked Cop on Adj-layer. 搭配形容词目标的系动词，在Adj词性层标记Cop。

Two adjective sub-types: ordinary descriptive adjectives, relational modifiers. 形容词分为普通描写形容词、关系修饰形容词。

Relational modifiers (economic, medical, military) cannot be used predicatively. Phrase-Type A (non-maximal adjectival). 关系修饰形容词（economic经济的、medical医学的、military军事的）不能作谓语；短语类型标记A（非最大形容词性成分）。

(135) The White House announced a new economic policy. (135) 白宫公布一项新的经济政策。

> economic is relational modifier, PT = A. economic是关系修饰形容词，短语类型A。

Predicative adjectives take complements: PP, Sfin, S-to-infinitive, Gerund-ing. 作谓语的形容词可以带各类补语：介词短语、限定小句、to不定式、-ing动名词。

(Phil is happy [that he passed] / [to finish] / [writing]). 菲尔很高兴【他通过了】/【完成】/【写作】。

3.6 Annotation with adverbs as targets

3.6 以副词作为标注目标

FrameNet mainly annotates speaker-attitude and epistemic-evidential adverbs as targets. FrameNet主要把说话人态度类、认识/证据类副词作为标注目标。

Manner adverbs (wisely, happily) are mostly annotated as FE dependents of verb/clause, not as target themselves. 方式副词wisely、happily大多作为动词、小句的从属框架元素标注，一般不作为标注目标。

(141) Frankly, my dear, I don't give a damn. (141) 坦白讲，亲爱的，我一点也不在乎。

> frankly is speaker-attitude adverb target. frankly说话人态度副词，作为标注目标。

Scope of adverb target may cover whole Sfin, or only embedded VP-to. 副词目标辖域可以覆盖整个限定小句，也可以只覆盖内嵌to不定式动词短语。

3.7 Annotation with prepositions as targets

3.7 以介词作为标注目标

Prepositions are targets primarily for spatial / temporal / motion frames (Locative relation frame). 介词主要在空间、时间、运动框架（方位关系）中充当标注目标。

Two core FEs for preposition targets: Figure (moving / located entity), Landmark (reference ground). 介词目标两个核心FE：Figure图形（运动/所处事物）、Landmark地标参照背景。

Grammatical Function: Figure → Ext external argument; Landmark → Obj object of preposition. 语法功能：Figure图形分配Ext外部论元；Landmark地标分配Obj介词宾语。

Copula be is marked Cop on Prep-layer for locative-relation sentences. 方位关系句子中，系动词be在Prep介词层标记Cop。

(148) [Cinderella Figure] was in [the cupboard Landmark]. (148) [图形 灰姑娘]处在[地标 橱柜]里面。

Modifiers to preposition get GF Dep, appropriate AVP / QUANT PT. 介词的修饰成分分配GF Dep从属成分，搭配AVP副词短语或QUANT量词短语。

(153) New Oxford is [10 miles QUANT] before Gettysburg. (153) 新牛津距离葛底斯堡还有【QUANT 十英里】。

3.8 Annotation with conjunctions as targets

3.8 以连词作为标注目标

Conjunction targets are rare, mostly for conditional, causal, temporal-relation frames. 连词作为标注目标很少见，主要用于条件、因果、时间关系框架。

If in Conditional_occurrence frame is typical example. Conditional_occurrence条件发生框架中的if是典型例子。

One clause gets GF Dep (complement of conjunction), the other main clause gets GF Head. 一个小句分配GF Dep连词补语；另一个主句分配GF Head中心。

(154) If [you want peace Dep] [you should give back the land Head]. (154) 如果【Dep 你想要和平】，【Head 你应当归还土地】。

3.9 Annotation relative to slot-fillers

3.9 针对槽填充项的标注

Slot-filler annotation (Gov-X annotation): for artifact / natural-kind nouns that do not evoke their own frame. 槽填充者标注（Gov-X）：针对本身不唤起独立框架的人造物、自然类名词。

The noun only fills FE slot of some other frame evoked by its governor verb. 该名词只充当支配动词所唤起别的框架的框架元素槽填充项。

Governor verb is tagged Gov; the artifact-noun headed constituent marked X. 支配动词标记Gov；人造物名词作中心词的成分标记X。

(156) He [built Gov] [two tall towers X]. (156) 他【Gov 建造】【X 两座高塔】。

Transparent nouns (cup of water, a sliver of bread) also use Gov-X annotation. 透明名词（a cup of water一杯水、a sliver of bread一小片面包）同样使用Gov-X标注。

(165) Leslie [ate Gov] [a [sliver Piece] [of lemon Substance]X]. (165) 莱斯利【Gov 吃了】【X 【Piece 小片】【Substance 柠檬】】。

> Here sliver is transparent quantifier-like noun, lemon is real substance argument of eat. sliver是类似量词的透明名词；lemon才是eat真正的物质论元。

Chapter 4 Identifying Phrase Types

第4章 识别短语类型

The FrameNet phrase-type inventory is built for lexicographic description purpose, not for full-blown general syntactic theory. FrameNet短语类型集合是为词典描写构建，不是服务于完备通用句法理论。

Only label constituents relevant to target predicate's valence needs. 只标记和目标谓词配价需求相关的成分。

4.1 List of phrase types

4.1 短语类型总列表

4.1.1 Noun Phrase Types

4.1.1 名词短语类型

- Non-referential NP (expletive it / there): no FE / GF / PT assigned; mark Null on Other-layer.
- 无指称名词短语（虚设it、there）：不分配FE、GF、PT；Other层标记Null。
- Poss(Possessive NP): possessive determiners, 's-genitive NPs, GF = Gen.
- Poss领属名词短语：领属限定词、's属格名词短语，语法功能Gen属格。
- N Non-maximal Nominal: incomplete nominal inside compounds / modification; cannot stand alone as argument.
- N非最大名词性成分：复合词内部、修饰位置的不完整名词，不能独立充当论元。
- NP Standard Noun Phrase: fully referential complete nominal argument.
- NP标准名词短语：具备完整指称能力的完整名词论元。

4.1.2 Prepositional Phrase Types

4.1.2 介词短语类型

- PP ordinary PP / particle (particle has no object).
- PP普通介词短语 /小品词（小品词无宾语）。
- PPing PP taking gerund-ing complement.
- PPing：-ing动名词作补语的介词短语。
- PPinterrog PP taking wh-clause complement.
- PPinterrog：wh疑问小句作补语的介词短语。
- PPadjP PP taking adjective-phrase complement.
- PPadjP：形容词短语作补语的介词短语。

4.1.3 Verb Phrase Types

4.1.3 动词短语类型

- VPfn Finite Verb Phrase.
- VPfn限定动词短语。
- VPbrst Bare-stem infinitive.
- VPbrst裸词干不定式。
- VPto To-marked infinitive.
- VPto to标记不定式。
- VPed Past-participle participial phrase.
- VPed过去分词短语。
- VPing Gerundive / present participle phrase.
- VPing动名词/现在分词短语。

4.1.4 Clause Types

4.1.4 小句类型

- Sfin Finite declarative complement-clause.
- Sfin限定陈述补语小句。
- Sinterrog Wh-interrogative clause.
- Sinterrog wh疑问小句。
- Swhether Whether / if interrogative clause.
- Swhether whether / if疑问小句。
- Sing Gerundive clause.
- Sing动名词小句。
- Sto To-marked subordinate clause.
- Sto to标记从属小句。
- Sforto For-to-marked subordinate clause.
- Sforto for-to标记从属小句。
- Sbrst Bare-stem subordinate clause.
- Sbrst裸词干从属小句。
- Sabs Absolutive non-finite depictive clause.
- Sabs绝对式非限定描绘小句。
- Srel Finite relative clause.

- Srel限定关系小句。
- Sub General subordinate clause introduced by subordinating conjunction.
- Sub由从属连词引导的通用从属小句。

4.1.6 Adjective Phrase Types

4.1.6 形容词短语类型

-A Non-maximal Adjectival (relational modifier inside compound / prenominal modification). -A非最大形容词性成分（复合词内部、名词前位置的关系修饰成分）。 -AJP Standard full adjective phrase. -AJP标准完整形容词短语。

4.1.7 Adverb Phrase

4.1.7 副词短语

-AVP Adverb Phrase. -AVP副词短语。

4.1.8 Quantifier Phrase

4.1.8 量词短语

-QUANT Quantifier-phrase for numerals / quant determiners. -QUANT量词短语，用于数词、量词限定词。

4.1.9 Quote

4.1.9 引语

-QUO Direct quote / inner monologue fragment. -QUO直接引语、内心独白片段。

> Rest of chapter 4 (4.2-4.9): detailed rules for assigning each PT label; examples of discontinuous PT, complex preposition, preposition-stranding, heavy shift etc. 第4章剩余4.2-4.9节：分配各个短语类型标签的详细规则；非连续成分、复杂介词、介词悬置、重成分移位等案例。

Chapter 5 Assigning Grammatical Functions

第5章 分配语法功能

Grammatical Function (GF) labels describe how a constituent satisfies the target predicate's valence requirements; not surface position. 语法功能GF描述成分如何满足目标谓词配价要求，不等于表层句法位置。

Main GF inventory: 主要语法功能清单：

1. Ext External Argument: corresponds to subject-like argument of target predicate.
2. Ext外部论元：对应目标谓词类主语论元。
3. Obj Object: corresponds to object-like argument of target predicate.
4. Obj宾语：对应目标谓词类宾语论元。
5. Dep Dependent: everything else; no distinction between complement vs adjunct.
6. Dep从属成分：其余全部成分；不再区分补语与附加语。
7. Gen Genitive: for possessive determiners.

8. Gen属格：领属限定词。
9. Appos Appositive: for noun-target appositive constituents.
10. Appos同位语：名词目标的同位成分。
11. Head: for adjectives / adverbs, the modified head item.
12. Head中心：形容词、副词场景，被修饰的中心成分。

> Important: FrameNet does NOT distinguish complement vs adjunct by GF labels; this distinction is captured by FE coreness status (core / peripheral / extra-thematic). 重要：FrameNet不依靠GF区分补语/附加语；该区分由FE的核心性状态实现（核心、外围、非题元）。

5.2 Assigning GFs for Verbs

5.2 动词的语法功能分配

-Ext: finite-clause subject; subject of raising / control predicate interpreted as target-verb argument. -Ext：限定小句主语；提升/控制谓词的主语被解读为目标动词论元。 -Obj: direct object; wh-extracted object; object-control embedded subject. -Obj：直接宾语；wh移位提取宾语；宾语控制结构内嵌主语。 -Dep: all other dependents (PP, adverb, subordinate clause etc). -Dep：其余全部从属成分（介词短语、副词、从属小句等）。

5.3 Assigning GFs for Adjectives

5.3 形容词的语法功能分配

-Ext: the argument that the adjective predicates over. -Ext：形容词所述谓的论元。 -Head: prenominal-adjective modified head noun. -Head：前置形容词所修饰的中心名词。 -Dep: PP complements, adverb modifiers, subordinate-clause complements. -Dep：介词短语补语、副词修饰语、小句补语。

5.4 Assigning GFs for Adverbs

5.4 副词的语法功能分配

-Ext: entity / situation that adverb predicates of. -Ext：副词述谓的实体或者事态。 -Head: the clause / VP modified by adverb. -Head：副词所修饰的小句、动词短语。 -Dep: adverb-internal modifiers. -Dep：副词内部修饰成分。

5.5 Assigning GFs for Prepositions

5.5 介词的语法功能分配

-Ext: Figure argument (the entity located / moving). -Ext：Figure图形论元（所处、运动的实体）。 -Obj: preposition object = Landmark reference-ground. -Obj：介词宾语，即Landmark地标参照背景。 -Dep: modifiers of preposition. -Dep：介词的修饰成分。

Chapter 6 Semantic Relations and Types

第6章 语义关系与语义类型

Frame-to-frame semantic relations defined in FrameNet: FrameNet定义的框架-框架语义关系：

1. **Inheritance(继承)** IS-A子类型关系；子框架FE绑定父框架对应FE。

2. **Perspective_on(视角)**：子框架从特定视角看待中性父背景框架。
3. **Using(使用)**：子框架预设父框架作为背景，不需要全部FE一一绑定。
4. **Subframe(子框架)**：子框架代表父复杂事件内部的子事件。
5. **Precedes(先于)**：一个框架事件时间上发生在另一框架事件之前。
6. **Causative_of(致使)**：使成框架 ↔ 起始框架配对关联。
7. **Metaphor(隐喻)**：隐喻映射关联。
8. **See_also(参见)**：松散参考链接。

6.2 Semantic Types

6.2 语义类型

Semantic Types are tags attached to frames, FEs or lexical-units to encode ontological / lexical / spatial properties. 语义类型是附加在框架、FE、词汇单元上的标签，编码本体、词汇、空间属性。

Categories: 分类：

1. Ontological types (entity, event, state, abstract-object etc.)
2. 本体类型（实体、事件、状态、抽象客体等）
3. Framal-type (properties specific to frame-semantic constructs)
4. 框架类型（框架语义构式专属属性）
5. Lexical-type (register, evaluative polarity, part-of-speech etc.)
6. 词汇类型（语体、评价极性、词性等）
7. Spatial semantic-types (motion-mode, spatial-shape etc.)
8. 空间语义类型（运动模式、空间形态等）

6.3 How to Use Relations

6.3 如何使用框架关系

-Look-up related frames / lexical units via frame-relation network. -依托框架关系网络，查询关联框架、关联词汇单元。 -Compare FrameNet representation against WordNet and other semantic-annotation systems. -把FrameNet表征和WordNet以及其他语义标注系统做对比。

Appendix A Major Extra-thematic Frame Elements

附录A 主要非题元框架元素

List of frequently-used extra-thematic FEs: Iteration, Particular_iteration, Point-of-contact, Recipient, Depictive, Event-description, Reencoding, Containing_event, Coordinated_event, Correlated-variable, Epistemic-construction etc. 高频非题元FE清单：Iteration迭代、Particular_iteration特定迭代、Point-of-contact接触点、Recipient接收者、Depictive描绘成分、Event-description事件描述、Reencoding重编码、Containing_event包含事件、Coordinated_event协同事件、Correlated-variable关联变量、Epistemic-construction认识类构式等。

Second-layer annotation for context-induced interpretations. 针对语境推导解读的第二层标注规则。

Appendix B Consistency Management in FrameNet

附录B FrameNet内部一致性管理

-Technical architecture of FrameNet database. -FrameNet数据库技术架构。 -Achievements of FrameNet project. -FrameNet项目已取得成果。

Bibliography

参考文献

Frame Semantics, FrameNet related cited publications list. 框架语义学、FrameNet相关引用文献列表。

Index

索引

Subject index for terms, frame-names, FE-names, lexical-units. 术语、框架名、框架元素名、词汇单元主题索引。